Building a Design-Based Matching Pipeline: From Principles to Practical Implementation in R

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Abstract

Matching, a canonical design for observational studies, takes many forms that rest on distinct — yet often implicit — statistical principles. We construct a matching pipeline for practitioners that makes these principles explicit and integrates stages often treated separately within a coherent design-based framework. The pipeline begins from the conceptual ideal of a randomized experiment, traces how observational studies depart from it, and then employs matching to approximate that ideal. The next stage is inference under the as-if randomization assumption that matched sets are equivalent to a collection of randomized experiments within blocks, where each block has a fixed number of treated units equal to the number observed in that set. Under this assumption, we consider inference on all individual effects in the "sharp" framework and on the average effect in the "weak" framework. The final stage is a sensitivity analysis to assess, under either framework, how inferences change under departures from as-if randomization. Each step includes extensively commented `R` code that equips practitioners to implement both established and newly developed procedures, including several not yet available in existing `R` packages. We illustrate the full workflow through an application examining the effect of United Nations peacekeeping interventions on the duration of post-conflict peace.

1 Design-Based Foundations of the Matching Pipeline

1.1 The Randomized Experimental Ideal

Imagine a randomized experiment in which a researcher flips a fair coin independently for each individual in the study. Heads means assignment of the individual to control, while tails means assignment to treatment. After assignments to treatment and control, the researcher administers the conditions and then compares outcomes between treated and control groups.

Why is this procedure effective? Randomization is a *fair lottery*, in that every individual has the same chance of being assigned to treatment. This means that individuals who would respond more strongly to treatment are no more likely to receive it than those who would respond less strongly, and the same is true for control. Because each individual's assignment is determined by the same coin toss (with the same probability of landing heads or tails), randomization leaves only two possibilities: (1) the difference in outcomes between treatment and control groups reflects the true causal effect, or (2) chance variation produced a misleading difference. Although misleading differences can occur by chance, randomization is valuable because it enables us to use statistical tools to quantify and limit the chance of such errors. As a result, randomized experiments yield especially credible causal conclusions.

Randomization is useful not only as a procedure, but also as an idea. It helps us understand when statistical tools will (and will not) yield credible conclusions, even when we have not directly randomized. In an observational study, the researcher does not control who receives the treatment and instead observes units after they have been assigned to treatment and control groups. In such settings, the idea of randomization can guide how we design studies so that they yield more credible causal conclusions.

1.2 Bridging Randomized Experiments and Observational Studies

A useful framework for connecting randomized experiments to observational studies is what Rubin (1977) calls "assignment to treatment group on the basis of a covariate," where a covariate is a pre-treatment characteristic of a study's units. To make this idea concrete, suppose treatment is assigned by independent coin tosses for each individual in which the probability of heads or tails depends on that individual's value of a single covariate. Consequently, all individuals with the same covariate value share the same chance of receiving treatment, while those with a different covariate value share a different chance.

In this setting, we can envision forming groups (i.e., matched sets) so that all individuals within a group share the same covariate value. Each group then functions as a miniature randomized experiment in that random chance alone explains why some individuals in the group received treatment while others did not. Importantly, to justify this interpretation, we do not need to know each individual's actual probability of treatment. It is enough to know that the probability of treatment depends only on the covariate, which ensures that all individuals with the same covariate value have the same chance of treatment.

1.3 Matching to Approximate the Randomized Experimental Ideal

The same intuition applies to matching when the probability of treatment depends on many baseline covariates. The underlying idea is that individuals with similar covariates have similar treatment assignment probabilities. Thus, in an effort to recreate a randomized experiment, we use the covariates we observe and believe determine treatment chances in order to divide individuals into groups, with each group containing both treated and control subjects who are homogeneous in those covariates. We are, in effect, constructing a new single variable: group (i.e., matched set) membership. We can think of this membership variable as the single covariate in the framework of "assignment to treatment group on the basis of a covariate" (Rubin, 1977) discussed in Section 1.2. Although we still do not know each individual's treatment probability, the hope is that all individuals within a group share the same probability, whatever it may be.

We call the condition in which all individuals within each matched set share the same treatment probability, strictly between 0 and 1, *as-if randomization.* As-if randomization is closely related to the conditions known as *selection on observables* and *strong ignorability*, although those conditions are stated in terms of covariates rather than matched sets. The term as-if randomization is especially fitting because, when all units within each matched set share the same individual treatment probability, all assignments within each set — holding fixed the set's observed number of treated units — are equally likely (treating the matched sets themselves as fixed; see Section 3.1). In this case, the probability distribution over these assignments is equivalent to that induced by a randomized experiment in which a researcher first forms blocks corresponding to the matched sets and then randomly assigns a fixed number of units to treatment within each block. Thus, when as-if randomization holds, we can apply the same statistical tools that we would use under such a block-randomized design to draw credible causal inferences from the observational study.

1.4 Why Sensitivity Analysis Matters

Unlike randomized experiments, even the best matched designs rely on the strong assumption of as-if randomization. When this assumption holds, we can draw causal conclusions by analyzing the data as if they came from a randomized experiment. However, if the assumption is wrong, our causal claims are no longer guaranteed to be credible.

How can this assumption fail? First, individuals within a group may be similar on observed covariates, but not similar enough to have the same treatment probabilities. Second, treatment assignment may depend on covariates we did not measure. If so, even if individuals within groups appear comparable on observed covariates, those individuals may still differ on hidden covariates that determine the probability of treatment.

For these reasons, it is important to assess the sensitivity of our causal conclusions to departures from as-if randomization. Conclusions are especially convincing when they hold not only under this assumption, but also under moderate violations of it. Conclusions that collapse under only mild departures are much less convincing.

1.5 Roadmap of the Design-Based Matching Pipeline

Building on these design-based foundations, we now outline a pipeline that starts with matching and then proceeds to inference and sensitivity analysis:

- **Construct and evaluate matched sets**. Given the specified covariates to be balanced, we begin with the mechanics of optimal matching (Hansen, 2004; Hansen and Klopfer, 2006): choosing a distance measure that defines similarity on the covariates, setting calipers — maximum allowable distances between treated and control units for inclusion in the same matched set — and imposing structural constraints (e.g., requiring matches to be pairs). We then show how to evaluate the resulting design in terms of both effective sample size and covariate balance. For the latter, we focus on tests proposed by Hansen and Bowers (2008) that compare the matched design's covariate balance to what one would expect under an equivalent completely randomized experiment within blocks (that is, random assignment with a fixed number of treated units per block).

- **Draw causal inferences**. Once a matched design is chosen, practitioners can conduct inference under as-if randomization, either within a “sharp” causal framework, which pertains to individual-level effects for all units, or a “weak” framework, which pertains to a summary quantity of the unit-level causal effects, typically the average treatment effect (ATE). For the sharp framework, we focus on how researchers can use both simulation- and Normal-based approximations to the exact randomization distribution to perform hypothesis tests, which can be inverted to obtain confidence sets and point estimates. For the weak framework, we cover estimation of the ATE and hypothesis tests about it. We emphasize exposition and code for recent variance estimators tailored to designs with only 1 treated or 1 control unit per matched set (Fogarty, 2018; Pashley and Miratrix, 2021) — an important case, since any matched design can be refined into sets that each contain one treated unit or one control unit without increasing covariate distances within sets (Gu and Rosenbaum, 1993; Rosenbaum, 1991; Hansen, 2004), although such refinement can come at the expense of effective sample size (Hansen, 2004, 2011).
- **Assess sensitivity**. Finally, we turn to sensitivity analyses under both inferential frameworks. We review established methods for conducting sensitivity analysis for tests of sharp nulls under possible violations of as-if randomization (Rosenbaum and Krieger, 1990; Gastwirth et al., 2000; Rosenbaum, 2018). We then describe and implement new methods that extend sensitivity analysis to tests of weak null hypotheses (Fogarty, 2023).

Figure 1 presents a flow diagram summarizing the overall pipeline, partitioned into three stages that correspond to Sections 2.2–2.4. The diagram displays each step and decision point, the relevant `R` tools (whether existing packages or custom functions included herein), and core references. All relevant code and replication materials are publicly available in the manuscript’s companion GitHub repository, https://github.com/tl2624/matching-guide.

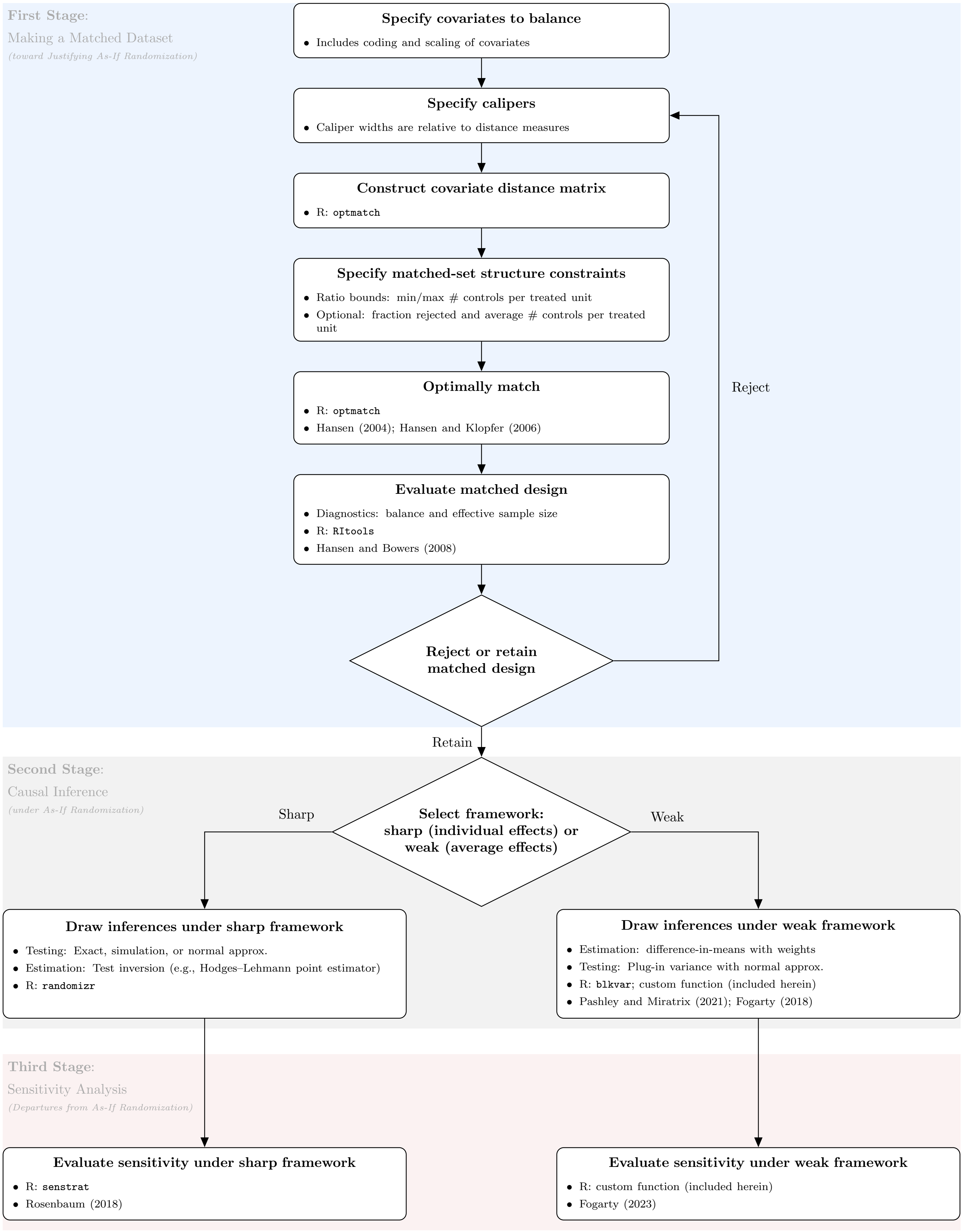
First Stage:
Making a Matched Dataset
(toward Justifying As-If Randomization)
Specify covariates to balance
• Includes coding and scaling of covariates
Specify calipers
• Caliper widths are relative to distance measures
Construct covariate distance matrix
• R: optmatch
Specify matched-set structure constraints
• Ratio bounds: min/max # controls per treated unit
• Optional: fraction rejected and average # controls per treated unit
Optimally match
• R: optmatch
• Hansen (2004); Hansen and Klopfer (2006)
Evaluate matched design
• Diagnostics: balance and effective sample size
• R: RItools
• Hansen and Bowers (2008)
Reject or retain matched design
Reject
Retain
Second Stage:
Causal Inference
(under As-If Randomization)
Select framework: sharp (individual effects) or weak (average effects)
Sharp
Weak
Draw inferences under sharp framework
• Testing: Exact, simulation, or normal approx.
• Estimation: Test inversion (e.g., Hodges–Lehmann point estimator)
• R: randomizr
Draw inferences under weak framework
• Estimation: difference-in-means with weights
• Testing: Plug-in variance with normal approx.
• R: blkvar; custom function (included herein)
• Pashley and Miratrix (2021); Fogarty (2018)
Third Stage:
Sensitivity Analysis
(Departures from As-If Randomization)
Evaluate sensitivity under sharp framework
• R: senstrat
• Rosenbaum (2018)
Evaluate sensitivity under weak framework
• R: custom function (included herein)
• Fogarty (2023)

Figure 1: Flow diagram of the design-based matching pipeline.

2 Implementation of the Matching Pipeline

We provide a high-level overview of the ideas behind matching and include code that demonstrates how to implement those ideas with various `R` packages. We also work through some of these steps "by hand" to underscore the underlying conceptual issues. Doing so also gives practitioners more flexibility to adapt the matching pipeline to their own needs.

We break the implementation into specific decision points that practitioners commonly face, and present the pipeline in three main stages (see Figure 1):

First Stage: Making a Matched Dataset (toward Justifying As-If Randomization)

(a) How Do I Measure Similarity on My Chosen Covariates?
  - Similarity on the Estimated Propensity Score

(b) How Can I Apply Rules for Matches to Ensure Comparability?
(c) How Can I Apply Rules for Matches to Improve Effective Sample Size?
(d) How Do I Actually Form the Matches?
(e) How Do I Decide Whether to Move Ahead with My Matched Design?

Second Stage: Causal Inference (under As-If Randomization)

(a) How Do I Draw Inferences under the Sharp Framework?
(b) How Do I Draw Inferences under the Weak Framework?

Third Stage: Sensitivity Analysis (Departures from As-If Randomization)

(a) How Do Inferences under the Sharp Framework Change under these Departures?
  - Finding the Worst-Case Scenario of Hidden Confounding for Valid Inference
    - Separable Approximation
    - Taylor Series Approximation
  - Conducting Worst-Case Sensitivity Analysis

(b) How Do Inferences under the Weak Framework Change under these Departures?

Before turning to First Stage, we introduce a running example taken from Gilligan and Sergenti (2008), which we use throughout this manuscript.

2.1 Running Example: United Nations Peacekeeping and Post-Conflict Peace

We introduce matching through an example that examines the causal effect of United Nations (UN) peacekeeping missions on the durability of post-conflict peace, a question of central importance for both academic research and policy. Our example draws on data from Gilligan and Sergenti (2008), whose title includes the phrase "Matching to Improve Causal Inference," underscoring the value of applying matching to study the UN's causal impact on post-conflict peace. These data are publicly available in the supplementary information of the article's webpage in the *Quarterly Journal of Political Science* (DOI: 10.1561/100.00007051).

The dataset we use from Gilligan and Sergenti (2008) includes 87 observations, each corresponding to a country's peace period episode following a civil war, with episodes beginning as early as January 1989 and data extending through December 2003. In some episodes, UN peacekeepers intervened (e.g., Sierra Leone, Jan 2001 – Dec 2003), while in others they did not (e.g., Macedonia, Sep 2001 – Dec 2003). The treatment variable is UN intervention (`UN`), coded as 1 if a UN mission was present during the peace period and 0 otherwise. The outcome variable is the duration of the peace spell, which Gilligan and Sergenti (2008) measure as the log of the number of months from the start of peace until either the outbreak of a new conflict or right-censoring at December 2003. This log-transformed outcome (`ldur`) captures the total length of the peace period rather than the time elapsed after a potential UN intervention. In practice, however, UN interventions almost always began immediately after the onset of peace: "Of the 19 post-conflict UN interventions, the United Nations was present within the first month for 16 of them" (Gilligan and Sergenti, 2008, p. 118). In what follows, we set aside these measurement details, with one caveat about censoring. The outcome `ldur` is right-censored for 15 of the 19 UN episodes but for only 28 of the 68 other episodes, so our analyses concern the log of the observed, possibly censored, duration rather than the log of the full duration of peace.

As Gilligan and Sergenti (2008) state, "UN missions are not randomly assigned" (p. 89). Whether peacekeepers are present during a given post-conflict peace period presumably depends on baseline covariates measured by Gilligan and Sergenti (2008), including the logged number of deaths (`lwdeaths`), the duration of the previous war in months (`lwdurat`; despite its name, this variable is not logged), ethnic fractionalization (`ethfrac`, a 0–100 index for which dividing by 100 is intended to represent the probability that two randomly chosen individuals belong to different ethnic groups), logged population size (`pop`), and others. We implement matching using these same covariates, but emphasize that our exercise is expository and not intended as a replication of the original findings.

2.1.1 Loading the Data for the Running Example

To load the data, you could first download the replication files from the supplementary information on the article's webpage (DOI: 10.1561/100.00007051), save the files to your working directory, and then use a package such as `haven` to load the Stata (.dta) file, `peace_pre_match.dta`, into `R`. However, for our purposes, we recommend loading our pre-created .rds file (`peace_pre_match.rds`), in which Stata's monthly numeric dates have been converted to `R`'s year-month format and the geographic region indicators recoded into a single factor variable (`region`). The following command loads this pre-created dataset (`peace_pre_match.rds`) into the `R` environment as an object named `data`.

```r
# Define base URL for the Matching Guide GitHub repo
base_url <- "https://raw.githubusercontent.com/tl2624/matching-guide/main"

# Load the cleaned dataset
data <- readRDS(
  url(
    paste0(
      base_url, "/data/peace_pre_match.rds"
    )
  )
)
```

2.2 First Stage: Making a Matched Dataset (toward Justifying As-If Randomization)

In an observational setting, treatment is not assigned by the flip of a coin but depends on individuals' covariates. The goal of matching is to compare treated and control units that have the same chances of treatment based on those covariates — i.e., the same *propensity scores*. If we could observe propensity scores, it would be straightforward to compare treated and control units by matching them directly on their propensity scores. Because propensity scores are not directly observed, we instead aim to create a collection of matched sets that is *balanced* — meaning that treated and control observations are similar in their covariates.

Before turning to questions of covariate similarity and matching, it is important to note that substantive transformations of covariates play a central role, as they determine the inputs on which subsequent notions of similarity between units are based. These transformations reflect substantive judgments and are often among the most important decisions in practice. In this manuscript, however, we do not address these substantive choices. Instead, we take the transformations used by Gilligan and Sergenti (2008) as given. For example, they measure ethnic fractionalization (`ethfrac`) on a 0–100 scale (rather than a 0–1 scale) and apply logarithmic transformations to covariates such as the number of deaths (`lwdeaths`) and population size (`pop`), among others. Our focus is on design and analysis decisions conditional on these substantively chosen scales.

2.2.1 How Do I Measure Similarity on My Chosen Covariates?

In the simplest terms, matching is about ensuring apples-to-apples, rather than apples-to-oranges, comparisons between treated and control observations (Rubin and Waterman, 2006). To create matched sets in which treated and control groups are similar in their covariates, we first need a distance measure that quantifies how close any two observations are. With such a measure in hand, we can then construct sets of treated and control observations that are close on this measure — i.e., apples-to-apples in their pre-treatment characteristics.

We record the distances between treated and control units in a distance matrix, in which the rows correspond to treated units and the columns to control units. Each cell of the matrix records the distance between a specific treated unit and a specific control unit, as defined by a distance measure. This distance measure takes the baseline covariates of the two units and maps them to a single nonnegative number, with smaller values indicating greater similarity.

In our setting, we are interested in similarity across 9 covariates, named in the object `covs`.

```r
# Define character vector of the 9 covariate names in the dataset
covs <- c("lwdeaths", "lwdurat", "ethfrac", "pop", "lmtnest", "milper",
          "bwgdp", "bwplty2", "region")
```

The first four — `lwdeaths`, `lwdurat`, `ethfrac`, and `pop` — were introduced earlier. The others include a logged measure of the percentage of a country's terrain that is mountainous (`lmtnest`), the logged total number of military personnel in a country (`milper`), logged GDP per capita before the last civil war (`bwgdp`), the Polity score (a -10 to 10 scale, from autocratic to democratic institutions) before the last civil war (`bwplty2`), and a region factor (`region`) with categories for Eastern Europe, Latin America, Asia, Sub-Saharan Africa, and North Africa/Middle East.

All of these covariates are measured before treatment and presumably determine the chance of a UN intervention during a country's peace period. Our interest in them stems primarily from their role in determining those intervention probabilities. Yet many of these covariates may also be prognostic; that is, they help predict the outcome of interest in Gilligan and Sergenti (2008), the log duration of the peace period (`ldur`) that countries would potentially experience with or without a UN intervention. This prognostic value of covariates can provide an additional reason to match on them (Hansen, 2008a; Sales et al., 2018).

There are many ways to measure the distance between a treated and a control unit. For example, we might compare units using the Euclidean distance — i.e., the square root of the sum of squared differences — across all baseline covariates. To do so, we first convert the factor variable `region`, which stores categorical labels, into a set of dummy (0 or 1) variables for each region. This conversion ensures that distances between treated and control units can be computed, since distance measures require numeric variables rather than categorical labels.

```r
# Install "dplyr" package (only run if not already installed)
# install.packages("dplyr")

# Load package for data manipulation (mutate, group_by, summarize, etc.)
library(dplyr)

# Convert categorical variable "region" into 0/1 dummy indicators
data <- data |> # Pipe (|>) to pass left-hand result into next function call
  mutate(
    # Use mutate() to create new variables by transforming existing columns
    # ifelse() returns one value if condition is TRUE and another if FALSE
    eeurop   = ifelse(test = region == "eeurop",   yes = 1, no = 0),
    lamerica = ifelse(test = region == "lamerica", yes = 1, no = 0),
    asia     = ifelse(test = region == "asia",     yes = 1, no = 0),
    ssafrica = ifelse(test = region == "ssafrica", yes = 1, no = 0),
    nafrme   = ifelse(test = region == "nafrme",   yes = 1, no = 0)
  )

# Covariate names with "region" replaced by dummy indicators
expanded_covs <- c(
  # setdiff() returns elements in 'x' that are not in 'y'
  setdiff(
    x = covs,
    y = "region" # Drops "region" from the covariate list
  ),
  "eeurop", "lamerica", "asia", "ssafrica", "nafrme"
)
```

As an example, we calculate the Euclidean distance between post-conflict Liberia, where the UN did intervene, and post-conflict Guinea-Bissau, where the UN did not.

```
# Extract covariates for Liberia and Guinea-Bissau (cname = country name)
liberia <- data[data$cname == "Liberia", expanded_covs]

guinea_bissau <- data[data$cname == "Guinea-Bissau", expanded_covs]

# Compute Euclidean distance between the two countries on these covariates
sqrt(
  sum(
    (liberia - guinea_bissau)^2
    )
  )  # Display the Euclidean distance
```

```
[1] 57.44263
```

We can obtain the full matrix of pairwise distance values using `match_on()` from the `optmatch` package. We do not need to manually recode factor variables into dummy indicators because `match_on()` handles this conversion automatically. (The function `match_on()` scales each region dummy by $1/\sqrt{2}$, so units in different regions differ by 1 rather than 2 in squared distance; distances within a region, including all entries in the submatrix that follows, equal the manual calculation.)

```
# Create a formula: UN (treatment indicator) ~ covariates
# Note: we keep "region" in covs as a factor
cov_fmla <- reformulate(
  termlabels = covs,
  response = "UN"
)
```

```r
# Install optmatch if not already installed
# install.packages("optmatch")

# Load optmatch, which provides the match_on() function
library(optmatch)

# Euclidean distance matrix: treated (UN = 1) vs control (UN = 0)
dist_mat_euc <- match_on(
  x = cov_fmla,                # Formula for covariates
  data = data,                 # Dataset used
  standardization.scale = NULL, # No rescaling of covariates
  method = "euclidean"         # Use Euclidean distance
)

# Add country names (cname) as row/column labels for clarity
dimnames(dist_mat_euc) <- list(
  data$cname[data$UN == 1], data$cname[data$UN == 0]
)

# Display submatrix of distances:
round(        # Round distances for display
  # treated units in rows 11-15 vs. control units in columns 27-30
  x = dist_mat_euc[11:15, 27:30],
  digits = 2  # Number of decimal places to round
)
```

```
              Niger Guinea   Togo Central African Republic
Sierra Leone  94.21 101.18 116.32                   116.94
Zaire         26.68  29.87  45.63                    46.57
Rwanda        66.75  71.57  77.04                    76.03
Mozambique   133.47 140.60 155.40                   155.80
Namibia      246.44 253.23 268.35                   268.08
```

In this subset of the full distance matrix, the first entry is the Euclidean distance of Sierra Leone (treated) from Niger (control). The last listed entry is the distance between Namibia (treated) and the Central African Republic (control).

One concern with Euclidean distance is that it depends on the scale of the covariates. For example, a country in Sub-Saharan Africa and a country in Latin America differ by 1 on each of two region indicators (`ssafrica` and `lamerica`), and these two differences together add 2 to the squared Euclidean distance; a difference of only about 1.4 months in the duration of the previous war (`lwdurat`) adds the same amount. Intuitively, we would not want such a tiny difference in war duration to be treated as equally important as belonging to different regions of the world. More generally, we want differences across variables to be placed on a comparable scale, so that a meaningful difference in one variable counts about the same as a difference of similar importance in another.

Another concern with Euclidean distance is that it ignores correlations among covariates. For example, countries with larger populations (`pop`) usually have more military personnel (`milper`), if only because a larger population provides a greater pool of potential recruits. Therefore, differences in both covariates (`pop` and `milper`) may largely reflect the same underlying factor — population size (`pop`). Yet Euclidean distance adds these differences separately, as if they were unrelated, which can exaggerate the overall distance between two observations.

The Mahalanobis distance (Mahalanobis, 1936) addresses both of these concerns. First, it standardizes covariates so that differences are placed on a comparable scale. Second, the Mahalanobis distance adjusts for correlations among covariates, ensuring that highly related variables are not effectively counted twice. We can construct a distance matrix based on Mahalanobis rather than Euclidean distances as follows.

```r
# Mahalanobis distance matrix: treated (UN = 1) vs control (UN = 0)
dist_mat_mah <- match_on(
  x = cov_fmla,
  data = data,
  standardization.scale = NULL,
  method = "mahalanobis" # Use Mahalanobis distance
)
```

The standardization in the Mahalanobis distance is distinct from researchers' substantive choices about how to transform covariates. Substantive transformations — such as logging certain covariates — reflect subject-matter considerations and precede the matching procedure. The standardization in the Mahalanobis distance is a statistical device that operates on the pre-specified covariates to compute distances comparably across covariates. The purpose of this device is to facilitate matches that are similar on the original covariate scales.

2.2.1.1 Similarity on the Estimated Propensity Score So far, we have focused on ways to measure distance between treated and control units across many covariates in order to identify which units are most similar and group them together to achieve covariate balance. However, when there are many covariates, it becomes difficult to find treated and control units that are similar on all covariates. This challenge is often referred to as the "curse of dimensionality."

A common way to address this problem is to reduce the information from many covariates into a lower-dimensional form. The *estimated propensity score* does this by collapsing information from all covariates into a single number. This number represents a transformation of an estimated linear index of covariates that accounts for how strongly each covariate predicts treatment.

Consider, for example, a logistic model for the estimated propensity score of an individual unit i. We write this model as

$$\hat{\lambda}(\boldsymbol{x}_i) = \frac{1}{1 + \exp(-\hat{\boldsymbol{\beta}}^\top \boldsymbol{x}_i)}, \tag{1}$$

where $\boldsymbol{x}_i$ is the covariate vector, $\hat{\boldsymbol{\beta}}$ is the vector of estimated coefficients, and the superscript $^\top$ denotes the transpose, turning the column vector $\hat{\boldsymbol{\beta}}$ into a row vector, so that $\hat{\boldsymbol{\beta}}^\top \boldsymbol{x}_i$ is a scalar. The quantity $\hat{\boldsymbol{\beta}}^\top \boldsymbol{x}_i$ is the estimated linear index, and the logistic (inverse-logit) function, $1/(1 + \exp(-x))$, maps any real-valued input, x, onto the interval (0, 1). The estimated linear covariate index for unit i, $\hat{\boldsymbol{\beta}}^\top \boldsymbol{x}_i$, is simply the logit, i.e., log odds, transformation of $\hat{\lambda}(\boldsymbol{x}_i)$ in (1).

This quantity, $\hat{\lambda}(\boldsymbol{x}_i)$, is a simple transformation of a linear index of covariates that best "separates" treated from control units. The estimated coefficients ($\hat{\boldsymbol{\beta}}$) "separate" treated from control units on the linear index because the coefficients are chosen to maximize the likelihood, which is larger when the fitted probabilities are close to 1 for treated units and close to 0 for controls. When treated and control units have little or no covariate overlap, the estimated linear indices can diverge substantially, very positive for treated units and very negative for controls. With greater overlap, the estimated linear indices for treated and control units are similar, clustering around a common value.

This estimation process reflects how predictive each covariate is of treatment, holding the other covariates fixed. A covariate that strongly predicts treatment, given the other covariates, receives a coefficient that turns typical differences in that covariate into large differences in the estimated linear index. A covariate with little predictive power receives a coefficient that turns typical differences in that covariate into small differences in the estimated linear index. Consequently, when we assess similarity on the estimated linear index of covariates in (1), the estimated coefficients assign greater importance to covariates that strongly predict treatment and less importance to those that do not.

To see this logic in action, we first estimate a logistic propensity score model using all covariates except `region`, which we exclude because some regions perfectly — or nearly perfectly — predict treatment assignment, leading to (near-)complete separation (Albert and Anderson, 1984).

```r
# Formula for UN ~ covariates (excluding "region")
psm_cov_fmla <- reformulate(
  termlabels = setdiff(
    x = covs,
    y = "region"
  ),
  response = "UN"
)

# Fit logistic regression for propensity score model
psm <- glm(
  formula = psm_cov_fmla,            # Treatment ~ covariates
  family = binomial(link = "logit"), # Logistic regression (logit link)
  data = data
)
```

Using this fitted propensity score model, we can extract each unit's estimated linear covariate index, $\hat{\boldsymbol{\beta}}^{\top}\boldsymbol{x}_i$, as follows.

```r
# Extract logit propensity scores (linear predictors from fitted model)
lin_cov_inds <- psm$linear.predictors  # = model.matrix(psm) %*% coef(psm)
```

The following code shows that the estimated linear covariate indices from the model `psm` correspond exactly to the logits (i.e., the log-odds transformations) of the model's predicted probabilities of treatment, which range between 0 and 1.

```r
# Extract estimated propensity scores (predicted probabilities of UN = 1)
p_scores <- psm$fitted.values

# Check that lin_cov_inds equals log odds (propensity scores on logit scale)
all.equal(
  lin_cov_inds, log(p_scores / (1 - p_scores))
)

# Also check that p_scores equals logistic(lin_cov_inds)
all.equal(
  p_scores, 1 / (1 + exp(-lin_cov_inds))
)
```

To see how two observations that differ greatly on a covariate can still have similar estimated propensity scores, consider post-conflict Haiti (treated; peace beginning in 1992) and Papua New Guinea (control; peace beginning in 1991). The two countries differ greatly in ethnic fractionalization (`ethfrac`), about 1 versus 45 on the 0–100 scale, but have the same logged number of battle deaths (`lwdeaths`) and similar logged numbers of military personnel (`milper`). Yet the estimated linear indices for Haiti and Papua New Guinea differ by only about 0.26. The gap in `ethfrac`, 43 points or about 1.7 standard deviations, raises Haiti's index by only about 0.13 because the estimated coefficient on `ethfrac` is close to 0 (about -0.003 per point). By contrast, the gap in `milper`, 0.29 or about 0.18 standard deviations, raises Haiti's index by about 0.26 because the estimated coefficient on `milper` is large (about -0.92). The estimated linear index therefore gives little weight to the covariate on which the two countries differ most and more weight to covariates that strongly predict treatment.

To illustrate these broader patterns, Figure 2 compares the distributions of the estimated linear covariate index for treated and control groups.

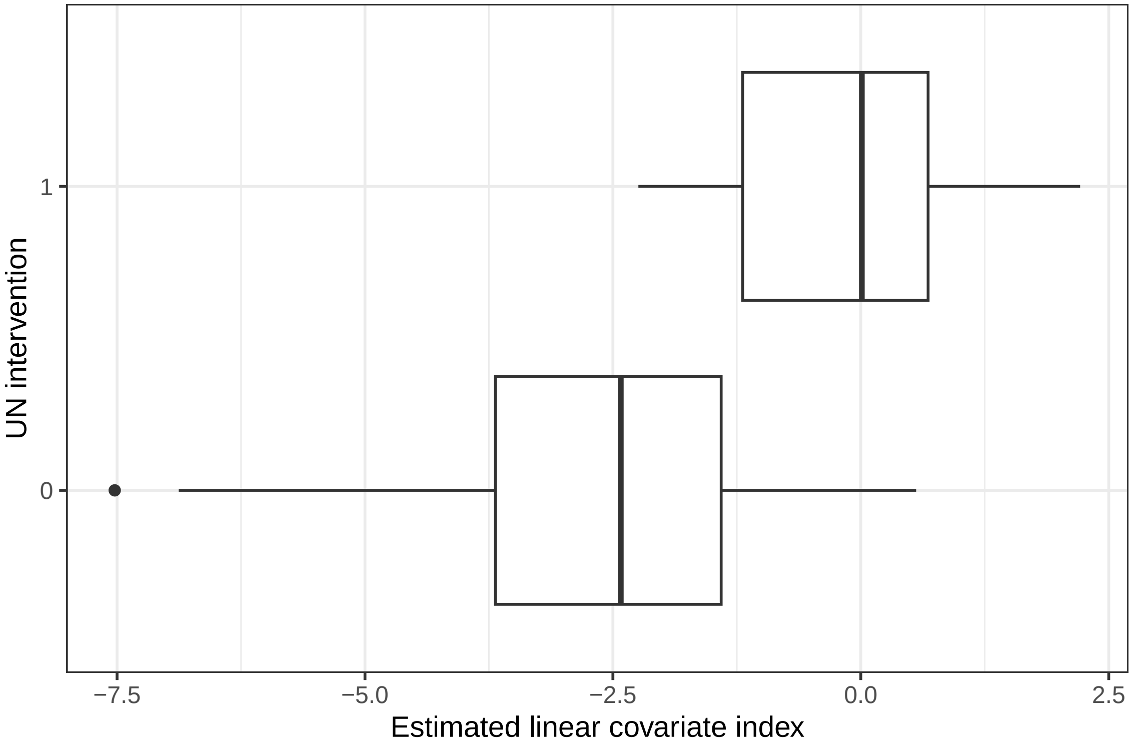

Figure 2: The empirical distributions of the estimated linear covariate index for treated and control groups.

As Figure 2 shows, there is some, but limited, overlap between treated and control groups. In accordance with our earlier discussion of how the estimated linear covariate index "separates" treated from control units, many control observations have very negative values (as low as -7.52), far from the treated units' range of -2.24 to 2.21. Nevertheless, a sufficient number of treated and control observations have estimated linear covariate indices that cluster around 0, indicating covariate overlap for at least a subset of treated and control observations.

2.2.2 How Can I Apply Rules for Matches to Ensure Comparability?

The discussion in Section 2.2.1 on measuring covariate distances between treated and control observations helps identify which treated–control pairs are similar. The goal of identifying these similar treated–control pairs is to form matched sets that are closely aligned on their covariates, thereby improving balance between treatment and control groups. In practice, we do this by excluding potential matches that are "too dissimilar." There are two common ways to do this:

- **Exact matching**: Require that units be identical on some subset of important covariates, typically those that are categorical or coarse enough for units to take the same values.
- **Calipers**: Impose a threshold for the maximum allowable distance so that no matched set may include a treated and a control unit that are farther apart than this caliper.

As a simple example to build intuition, we will impose the following constraints:

- Observations can belong to the same matched set only if they are in the same geographic region.
- Treated and control observations more than two points apart on the Polity score cannot be in the same matched set.

We now impose the first constraint, requiring an exact match on region. Doing so produces a distance structure, `em_region`, whose entry for a treated–control pair is 0 if the two units belong to the same region and ∞ (`Inf`) otherwise, so that treated and control units can be matched only within a region.

```r
# Create distance structure: 0 if units are in same region, Inf otherwise
em_region <- exactMatch(
  x = UN ~ region,
  data = data
)
```

Now we impose the second constraint by constructing a distance matrix based on the Polity score, `bwplty2`, with a caliper of 2. This matrix records the absolute difference in Polity scores when the difference is 2 or less, and assigns a value of ∞ (denoted in `R` as `Inf`) when the difference exceeds 2. The ∞ entries are crucial because `optmatch` minimizes the sum, across all matched sets, of the within-set sums of covariate distances between each treated-control pair. Consequently, any treated–control pair differing by more than 2 points on the Polity score is assigned an overall distance of ∞, and this infinite distance prevents the two units from being matched.

```r
# Euclidean distance on Polity score (bwplty2) with caliper = 2
# Pairs differing by > 2 are set to Inf
euc_dist_polity_cal_2 <- match_on(
  x = UN ~ bwplty2,
  caliper = 2,  # Set caliper
  data = data,
  standardization.scale = NULL,
  method = "euclidean"
)
```

Note that we used Euclidean distance here because, after exactly matching on geographic region, matching proceeds on only 1 covariate (Polity score), so we do not have to worry about covariates' relative scales.

Finally, as the following code shows, we combine the two constraints into distance matrices defined within each region by "adding" the objects `em_region` and `euc_dist_polity_cal_2`.

```r
# Overall distance matrix by element-wise addition of two distance matrices
overall_dist_mat <- em_region + euc_dist_polity_cal_2
```

2.2.3 How Can I Apply Rules for Matches to Improve Effective Sample Size?

Beyond comparability in covariates, we also care about the matched study's size. The effective sample size depends not only on the total number of units included in our matches but also on how those units are arranged across the matched sets.

The `optmatch` package will always produce matches with a particular arrangement of units across sets. In particular, all sets contain either 1 treated unit or 1 control unit — an overall structure that, absent ratio constraints, minimizes within-set covariate distances while discarding only units that have no permissible match (Rosenbaum, 1991; Gu and Rosenbaum, 1993; Hansen, 2004). In practice, optimal full matching can yield lopsided sets, with either 1 treated matched to many controls or 1 control matched to many treated units, which has implications for the effective sample size.

In the `optmatch` package, the effective sample size is defined as the sum, across matched sets, of the harmonic mean of the numbers of treated and control units within each set. The formal definition is given by

$$\text{Effective sample size} = \sum_{s=1}^{S} \left[\left(m_s^{-1} + (n_s - m_s)^{-1} \right) /2 \right]^{-1}, \tag{2}$$

where the index $s \in \{1, \ldots, S\}$ runs over the S matched sets, with m_s denoting the number of treated units in set s and $n_s - m_s$ the number of control units. With n_s denoting the number of units in set s, the total number of individuals included in the matched study is $n = \sum_{s=1}^{S} n_s$.

From the formula in (2), we can see how the effective sample size depends on the arrangement of units across sets. For example, in a study with 4 total units, the effective sample size would be 2 if the units were arranged into 2 matched pairs. By contrast, in a study of the same total size but arranged as a single set with 1 treated unit and 3 controls, the effective sample size would be 1.5. The effective sample size is larger in the former arrangement because it provides two distinct treated-versus-control comparisons, whereas the latter provides only one.

This definition of effective sample size anticipates subsequent outcome analysis under as-if randomization, as effective sample size bears directly on the precision of estimators and the power of hypothesis tests. Assuming constant, additive treatment effects within a set, the variance of the Difference-in-Means — the average outcome among treated units minus the average outcome among control units — in that set is minimized when the harmonic mean is largest (Hansen and Bowers, 2008; Hansen, 2011). For a fixed set size, the harmonic mean is largest when the numbers of treated and control units are equal. Thus, all else equal, matched pairs and other balanced sets provide more information about causal effects than lopsided sets with unequal treated-to-control ratios.

One straightforward way to increase effective sample size is to relax restrictions on which units can be matched. Relaxing calipers can admit additional units into the matched design by allowing matches for units that would otherwise be discarded. For instance, we might widen the caliper on Polity score from 2 to 3 and then rebuild the distance matrix.

```r
# Apply a caliper of width 3 to the Polity Euclidean distance matrix
euc_dist_polity_cal_3 <- match_on(
  x = UN ~ bwplty2,
  caliper = 3,
  data = data,
  standardization.scale = NULL,
  method = "euclidean"
)

# Combine regional exact match distance with Polity distance
em_region + euc_dist_polity_cal_3
```

Doing so increases the effective sample size, but such gains need not be substantial. When newly admitted units are absorbed into existing matched sets — each of which contains only one treated or one control unit — rather than generating additional treated-versus-control comparisons, the resulting contribution to effective sample size is limited. More generally, as the formula in (2) makes clear, the effective sample size depends on how units are distributed across sets. When units, including newly admitted ones, are concentrated in only a few sets, the resulting gain in effective sample size is modest.

Instead of relying solely on caliper width, researchers can also shape the effective sample size by controlling the structure of matched sets via the `min.controls` and `max.controls` arguments in `optmatch`'s `fullmatch()` function. These arguments specify lower and upper bounds, respectively, on the ratio of control to treated units within each matched set. By default, `min.controls = 0` and `max.controls = Inf`, which impose no restrictions on matched set composition. Changing these defaults allows researchers to control how balanced or lopsided matched sets may be, thereby shaping the effective sample size.

To illustrate, suppose we want to restrict matches using the `overall_dist_mat` introduced earlier. We know that `optmatch` will divide the data into matched sets containing one treated unit and any positive number of controls, or one control unit and any positive number of treated units. However, we can impose additional constraints on this full matching, such as requiring a minimum control-to-treated ratio of 1:2 — that is, at least one control for every two treated units (`min.controls = 0.5`) — and no more than two controls per treated unit (`max.controls = 2`). Under these restrictions, allowable matched sets could include 2 treated units with 1 control, 1 treated unit with 1 control (a matched pair), or 1 treated unit with 2 controls.

```
# Full matching using overall distance matrix
fullmatch(
  x = overall_dist_mat,
  min.controls = 0.5,    # At least 0.5 controls per treated unit
                         # (i.e., no more than 2 treated per control)
  max.controls = 2,      # At most 2 controls per treated unit
  omit.fraction = NULL, # Fraction of controls (or, if negative, treated) to discard
  mean.controls = NULL, # Governs average controls per treated unit
  # Only one of omit.fraction or mean.controls can be non-NULL
  data = data
)
```

Note, in addition, that the `fullmatch()` arguments of `omit.fraction` and `mean.controls` provide explicit levers for controlling how many treated and control units are discarded, though at most one of `omit.fraction` or `mean.controls` may be specified.

Imposing ratio constraints can have mixed consequences. In some cases, balance may worsen if a control is forced to match with a less similar treated unit — though still within the specified calipers — in order to satisfy the minimum and maximum ratio rules. In other cases, such constraints may improve effective sample size by redistributing how units are grouped without hurting balance. However, if the ratio constraints are too stringent, they can reduce effective sample size by forcing too many units to be discarded.

In applied settings, final choices of calipers and ratio constraints typically follow iterative checks of both covariate balance and effective sample size. Practitioners often compare several specifications and select among those that satisfy diagnostics for both goals. Hansen and Sales (2015) outline how to conduct this process in a structured way, drawing on the stepwise intersection–union principle (SIUP) of hypothesis testing.

2.2.4 How Do I Actually Form the Matches?

The simple matching example in Sections 2.2.2 and 2.2.3 — based on an exact match on geographic region and a caliper on Euclidean distance for a single covariate (Polity score) — serves to illustrate the basic ideas. When matching on many covariates, however, we will often prefer some combination of Mahalanobis distance and propensity score matching, sometimes adding calipers on specific covariates. In what follows, we use *rank-based* Mahalanobis distance, which has the advantage of being less sensitive to outliers and to heavily tied covariates, such as rare binary indicators (Rosenbaum, 2010, pp. 171–172). We further constrain the matching by imposing a caliper on the estimated propensity score, requiring treated and control observations to come from the same geographic region, and applying additional calipers directly to two covariates: ethnic fractionalization (`ethfrac`) and logged GDP per capita (`bwgdp`).

We impose a caliper equal to 0.5 standard deviations of the logit of the estimated propensity score (the estimated linear covariate index defined in (1)). This choice is larger than a common rule of thumb of 0.20 standard deviations. That rule of thumb traces to the finding of Cochran and Rubin (1973, Table 2.3.1, p. 421) that, for a single normally distributed covariate, a caliper of 0.20 standard deviations (the square root of the average of the treated and control variances) removes practically all of the bias. Given that the standard deviation of the logit index is approximately 1.96, our choice of 0.5 permits treated and control units to differ by up to roughly 0.98 on the logit scale, compared with about 0.39 under the 0.20 guideline. (With Cochran and Rubin's definition of the standard deviation, about 1.58 for the logit index here, the 0.20 guideline corresponds to about 0.32.)

```r
# Add logistic regression linear predictors to dataset
data$logit_p_score <- lin_cov_inds

# Population standard deviation of logit_p_score (divides by n, not n - 1)
pop_sd_logit <- sqrt(
  mean(
    (data$logit_p_score - mean(data$logit_p_score))^2
  )
)

# Distance matrix for logit of estimated propensity scores
ps_mat <- match_on(
  x = UN ~ logit_p_score,
  caliper = 0.5 * pop_sd_logit,
  data = data,
  standardization.scale = NULL,
  method = "euclidean"
)
```

We next construct the distance matrix for rank-based Mahalanobis distance.

```r
# Rank-based Mahalanobis distance on covariates
# covs defined earlier as covariate names; here, drop "region"
rank_mah_mat <- match_on(
  x      = reformulate(
    termlabels = setdiff(
      x = covs,
      y = "region"),
    response   = "UN"),
  data   = data,
  standardization.scale = NULL,
  method = "rank_mahalanobis" # Use rank-based Mahalanobis distance
)
```

Finally, we construct the Euclidean distance matrix for ethnic fractionalization (`ethfrac`) and logged GDP per capita (`bwgdp`) using calipers of 35 and 2, respectively. The exact-match constraint on region has already been defined through the object `em_region` in Section 2.2.2.

```
# Compute Euclidean distance matrix for ethnic fractionalization
eth_mat <- match_on(
  x = UN ~ ethfrac,
  caliper = 35,
  data = data,
  standardization.scale = NULL,
  method = "euclidean"
)

# Compute Euclidean distance matrix for logged GDP per capita
bwgdp_mat <- match_on(
  x = UN ~ bwgdp,
  caliper = 2,
  data = data,
  standardization.scale = NULL,
  method = "euclidean"
)
```

Next, we combine the `ps_mat`, `rank_mah_mat`, `eth_mat`, `bwgdp_mat`, and `em_region` objects to form the overall distance structure. We then pass the combined object to `optmatch`'s `fullmatch()` function, imposing a constraint that no more than 4 control units may be matched to any treated unit. If instead we wanted to perform pair matching, the `optmatch` package allows users to directly specify a matched-pair structure via the `pairmatch()` function. Alternatively, one can request pair matching by setting `min.controls = 1` and `max.controls = 1` in the `fullmatch()` call. The `pairmatch()` call and this `fullmatch()` call coincide when every treated unit can be paired, but `pairmatch()` additionally sets `omit.fraction` so as to require that all treated units be matched. The results of the two calls can therefore differ when calipers or exact-matching constraints make it impossible to match every treated unit. The following code chunk proceeds with full matching under the constraint that no more than 4 controls may be matched to any treated unit.

```
# Full matching:
# PS + rank Mahalanobis + Euclidean (ethfrac, bwgdp) + region exact
fm <- fullmatch(
# x specifies the distance structure:
# match_on() formula, distance matrix, or sum of match_on() distances
  x            = ps_mat + rank_mah_mat + eth_mat + bwgdp_mat + em_region,
  data         = data,
  max.controls = 4 # <= 4 controls per treated (min.controls = 0 by default)
)
```

To calculate the effective sample size of the matched observations, we use the following function from `optmatch`.

```
# Effective sample size of matched sets
effectiveSampleSize(
  fm
)
```

```
[1] 15.26667
```

Referring back to (2), this reported effective sample size of approximately 15.27 equals the sum across sets of the within-set harmonic mean of the number of treated and control subjects. This effective sample size reflects the matched structure in which no set contains more than 4 controls per treated observation.

```
# Summarize matched sets and report effective sample size
summary(
  fm
)
```

```
Structure of matched sets:
1:0 2:1 1:1 1:2 1:3 1:4 0:1
  6   1   4   4   2   1  45
Effective Sample Size:  15.3
(equivalent number of matched pairs).
```

The notation in the `summary()` output of `optmatch` indicates the ratio of treated to control observations within each matched set. For example, `1:0` indicates 1 treated unit and no controls (effectively an unmatched treated unit). Similarly, `1:2` indicates 1 treated unit and 2 controls, and `0:1` indicates no treated units and 1 control (effectively an unmatched control). For each label, the output shows how many matched sets have that particular structure. Here 6 of the 19 treated episodes are left unmatched, so the inferences that follow concern the 13 matched UN episodes and their 23 matched controls rather than all UN interventions.

If we examine the object returned by the matching call (`fm`), we see that `optmatch` labels each observation according to its matched set, assigning `NA` to those not included. Because we performed exact matching by region, `optmatch` labels each matched set using the name of the exact-match stratum followed by a set index within that stratum. For example, the label `lamerica.1` denotes the first matched set within the Latin America stratum.

To see which units were matched together, we can add the `fm` object to the data frame and then tabulate. For example, to view the `ssafrica.3` set, we run the following.

```r
# Add matched set ID to data for each unit
data$fm <- fm

# Look at one matched set ("ssafrica.3") for illustration
data |>
  filter(
    fm == "ssafrica.3" # Keep only units in set "ssafrica.3"
  ) |>
  # Display selected variables
  select(
    cname, UN, region, logit_p_score, ethfrac, bwgdp, bwplty2
  )
```

```
# A tibble: 4 x 7
  cname      UN region   logit_p_score ethfrac bwgdp bwplty2
  <chr>   <dbl> <fct>            <dbl>   <dbl> <dbl>   <dbl>
1 Burundi     0 ssafrica         0.508    3.55  5.35      -7
2 Rwanda      1 ssafrica         0.148   12.9   5.68      -7
3 Somalia     0 ssafrica         0.209    7.67  6.61      -7
4 Lesotho     0 ssafrica         0.558   22.2   6.28       0
```

We can see that the exact match on geographic region holds, since all 4 countries are located in Sub-Saharan Africa. The logit of the estimated propensity score is similar across units, though not identical. The treated country, Rwanda, is matched to three controls — Burundi, Somalia, and Lesotho — and in each case the distance falls within 0.5 standard deviations of the estimated logit propensity scores (approximately 0.98), the caliper we specified. Likewise, Rwanda's distances to each of the three control units fall within the calipers of 35 for ethnic fractionalization (`ethfrac`) and 2 for logged GDP per capita (`bwgdp`). By contrast, distances between control countries can technically exceed these thresholds, since `optmatch` enforces calipers only between treated and control units, not among controls.

We can also see that some covariates used in the propensity score model and the rank-based Mahalanobis distance — such as Polity score (`bwplty2`) — still show modest imbalance within this matched set. This imbalance is unsurprising. We applied a caliper on the logit of the estimated propensity score and matched on the rank-based Mahalanobis distance including `bwplty2`, but we did not apply a caliper directly to `bwplty2`, though such a caliper could easily be added if desired.

2.2.5 How Do I Decide Whether to Move Ahead with My Matched Design?

Once we have constructed our matched sets, we want to evaluate the overall quality of the matched design. Covariate balance is an important aspect of this evaluation. To assess covariate balance, we compare the balance in our matched observational study with the balance we would expect to see in a completely randomized experiment within blocks (Hansen and Bowers, 2008).

We next calculate adjusted covariate means for the treated and control groups by averaging set-specific means across matched sets, with each set weighted by its number of treated units, as `RItools` does by default; the adjusted treated mean is therefore the mean among matched treated units. We also report standardized differences, defined as the treated-minus-control difference in these adjusted means divided by a pooled standard deviation of the covariate — the square root of the average of the treated-group and control-group variances — computed in the overall data (including both matched and unmatched units). The significance tests, by contrast, weight each set by its contribution to the effective sample size (Hansen and Bowers, 2008).

```r
# Install "RItools" package (only run if not already installed)
# install.packages("RItools")

# Load RItools package for balance diagnostics (balanceTest)
library(RItools)
# Covariate balance test
cov_bal <- balanceTest(
  # Formula: treatment ~ covariates
  # update(): keep the original formula (. ~ .)
  # and add stratification by matched set, + strata(fm)
  fmla = update(
    cov_fmla, . ~ . + strata(fm)
  ),
  data = data,
  p.adjust.method = "none" # Method of p-value adjustment (none here)
)
```

	Before matching			**After matching**		
Covariate	Control mean	Treated mean	Std. diff	Control mean	Treated mean	Std. diff
Log Cumulative Battle Deaths from Last War	6.65	8.98	0.84*	8.34	8.57	0.08
Duration of Last War	50.28	80.53	0.39	60.51	73.23	0.16
Ethnic Fractionalization	56.50	49.21	−0.28	51.75	57.50	0.22
Log Population Size	9.51	8.75	−0.64*	8.89	8.91	0.02
Log Mountainous	2.22	2.80	0.43	2.69	2.82	0.09
Log Military Personnel	3.87	3.25	−0.42	3.63	3.54	−0.06
Log GDP per Capita Before Last War	6.56	6.59	0.03	6.73	6.55	−0.17
Democracy (Polity Score) Before Last War	−0.84	−2.58	−0.32	−2.19	−2.54	−0.07
Asia	0.19	0.00	−0.68*	0.00	0.00	0.00
Eastern Europe	0.15	0.37	0.51*	0.46	0.46	0.00
Latin America	0.12	0.21	0.25	0.08	0.08	0.00
North Africa & Middle East	0.12	0.11	−0.04	0.08	0.08	0.00
Sub-Saharan Africa	0.43	0.32	−0.23	0.38	0.38	0.00

Table 1: Adjusted covariate means for treated and control groups and standardized treated–control differences, before and after matching. Asterisks denote two-tailed p-values less than or equal to 0.05.

In Table 1, the adjusted means offer a direct description of balance. The standardized differences place all covariates on a common scale, enabling at-a-glance comparisons of imbalance across covariates. Stars indicate covariates for which the adjusted mean difference would be unusually extreme under a block-randomized experiment that assigns treatment completely at random within matched sets, holding fixed the observed number of treated units in each set. The stars are based on a Normal approximation to the randomization distribution of adjusted mean differences, and, because no adjustments are made for multiple comparisons, these stars are conservative for the purpose of judging balance. That is, although each covariate is tested at approximately the nominal level, the chance that at least one of the 13 covariates is starred under complete random assignment within matched sets exceeds that level, so the stars err toward flagging imbalance. Alternative adjustments for testing multiple covariates are available, such as the Holm procedure (Holm, 1979), which is the default option in `RItools`'s `balanceTest()` function.

One concern with balance tests is that high p-values may arise not from improved covariate balance but from the reduction in effective sample size that typically accompanies the matching process (Austin, 2008; Imai et al., 2008). As Hansen (2008b) notes, however, this possibility is less troubling than it first appears. The same increase in standard errors that produces high p-values for covariate balance tests will also affect subsequent causal inferences, meaning that those high p-values remain informative; they suggest that we are, if anything, less likely to overstate our causal conclusions than if the tests had been significant.

Regardless of whether the balance tests are statistically significant, there are also established guidelines for what constitutes sufficient balance. While precise thresholds depend on context and substantive judgment about each covariate's importance, two rules of thumb appear in Austin (2009) and Stuart (2010). Austin (2009) reports a proposal that standardized differences of 0.1 or greater indicate meaningful imbalance on a covariate, whereas Stuart (2010) gives a more lenient threshold of 0.25, attributing it to Rubin (2001). In our case, all covariates meet this latter standard, although three (duration of the last war, ethnic fractionalization and GDP per capita, with standardized differences of 0.16, 0.22 and -0.17) exceed the 0.1 threshold of Austin (2009); none show statistically significant differences, indicating that observed imbalances would not be unusual under a completely randomized experiment within blocks (i.e., under as-if randomization).

In addition to assessing balance on each covariate individually, Hansen and Bowers (2008) also propose an omnibus, chi-squared (χ^2) test that evaluates balance across all covariates and their linear combinations simultaneously. The following code conducts this test.

```
# Extract overall chi-square balance test by matched set (fm)
cov_bal$overall["fm", ]
```

```
   chisquare df   p.value
fm  2.631007  8 0.9553417
```

This χ^2 balance test yields an observed test statistic of 2.63 and a corresponding p-value of 0.96. The test has 8 rather than 13 degrees of freedom because its degrees of freedom equal the rank of the covariance matrix of the adjusted differences (Hansen and Bowers, 2008, p. 229). With exact matching on region, the five region indicators are constant within every matched set, so their adjusted differences are identically zero. Our high p-value indicates that the observed level of covariate balance is consistent with what we would expect in a completely randomized experiment within blocks.

Despite the high p-value, there is no guarantee that balance is sufficient for as-if randomization. Residual imbalance on observed covariates and hidden imbalance on unobserved ones may undermine the as-if randomization assumption. For now, we proceed under the as-if randomization assumption, but we will later assess how sensitive our inferences are to violations of it due to such imbalances.

2.3 Second Stage: Causal Inference (under As-If Randomization)

In both the sharp and weak frameworks, the causal targets of inference are defined through potential outcomes. Under what is known as SUTVA, the stable unit treatment value assumption (Rubin, 1980, 1986; Cox, 1958), each unit has two potential outcomes: a value the outcome would take if that unit were assigned to treatment and a value the outcome would take if that unit were assigned to control. Let $y_{si}(1)$ and $y_{si}(0)$ denote these potential outcomes, respectively, for unit i in set s, where the index i runs over the $\{1, \ldots, n_s\}$ units in set s. The individual treatment effect is $\tau_{si} = y_{si}(1) - y_{si}(0)$. With $n = \sum_{s=1}^{S} n_s$ total units, let $\boldsymbol{\tau} = (\tau_{1,1}, \tau_{1,2}, \ldots, \tau_{S,n_S})^\top$ collect all n unit-level effects, indexed first by matched set and then by unit within set, and write the ATE as $\tau = (1/n) \sum_{s=1}^{S} \sum_{i=1}^{n_s} \tau_{si}$.

Both $\boldsymbol{\tau}$ and τ are defined only for the subset of units retained after matching, which often differs from the original set of units prior to matching (Rosenbaum and Rubin, 1985; Rosenbaum, 2012). This difference is especially important when retention is driven by overlap in the estimated propensity score, as the resulting study population can be difficult to interpret substantively. In such settings, it is often preferable to define the study population directly in terms of a small number of substantively meaningful covariates, reflecting the argument in Rosenbaum (2010) that it is "usually better to go back to the covariates themselves … perhaps redefining the population under study to be a subpopulation of the original population" (p. 86), as also emphasized by Stuart (2010). Building on this insight, subsequent approaches provide formal procedures for defining the matched study population in this way (Fogarty et al., 2016; Traskin and Small, 2011).

The breadth of the study population retained after matching depends in part on the relative abundance of treated and control units across covariate space. When one treatment arm is sparsely represented in regions dominated by the other, overlap may be limited to a narrow region of covariate space and include relatively few units. All else equal, a larger pool of units available for matching supports overlap over a wider region of covariate space and a larger, more interpretable study population over which the target of inference is defined.

Neither target of inference, $\boldsymbol{\tau}$ nor τ, defined among the matched units, can be directly calculated. Even in a randomized experiment, we cannot assign a country emerging from civil war to receive a UN peacekeeping mission, observe how long the ensuing peace lasts, and then rewind time to observe how long peace would have lasted in the absence of a UN peacekeeping mission. In other words, for each unit we observe only one of its two potential outcomes. Let y_{si} denote this observed outcome, which equals the treated potential outcome when unit i in set s receives the treatment and the control potential outcome otherwise. Because the individual-level causal effect can never be directly observed — only one potential outcome is realized for each unit — we must rely on statistical inference to draw conclusions about $\boldsymbol{\tau}$ and τ.

We consider inference under the sharp and weak frameworks, targeting $\boldsymbol{\tau}$ and τ, respectively. A line of work shows how a single test can be exact for a sharp null and asymptotically valid for the corresponding weak null under as-if randomization (Chung and Romano, 2013; Wu and Ding, 2021; Cohen and Fogarty, 2022; Fogarty, 2023) (see also Ding, 2017 on conflicts between the two types of tests). The unification of sharp and weak tests does not generally extend to sensitivity analyses outside matched-pair designs (Fogarty, 2020a, 2023). When researchers must choose between the two, the decision depends on both statistical properties and substantive goals.

- Statistically, the sharp framework specifies all missing potential outcomes, allowing exact randomization inference under minimal assumptions. The weak framework leaves some outcomes unspecified and instead relies on variance estimation and a Normal approximation, which can perform poorly in small samples or when outcomes are skewed with extreme outliers. In such cases — or whenever one wants exact p-values under minimal assumptions — permutation inference under the sharp framework may be preferable.
- Substantively, researchers usually test a constant effect in the sharp framework. Such a hypothesis may be unrealistic or of limited scientific interest (see Gelman, 2011; Caughey et al., 2023). The weak framework, by contrast, accommodates heterogeneous effects across units, making the ATE a more relevant target in many settings. Nevertheless, tests of a constant effect can provide a useful approximation to a more complex hypothesis with heterogeneous effects (Rosenbaum, 2010, pp. 44–46), and such tests remain valid for a range of bounded but heterogeneous effects (Caughey et al., 2023).

2.3.0.1 The Assignment Process as the Basis for Inference Regardless of the framework, inference is based on the treatment assignment process. Let z_{si} denote an indicator for whether unit i in matched set s is treated ($z_{si} = 1$) or not ($z_{si} = 0$). We collect these indicators for all units in set s into the vector $\boldsymbol{z}_s = (z_{s1}, \ldots, z_{sn_s})^\top$. Stacking these vectors across all sets gives the full assignment vector $\boldsymbol{z} = (z_{11}, \ldots, z_{Sn_S})^\top$, again indexed first by matched set and then by unit within set. For inference, we condition on the number of treated units within each set, m_s, even if the actual assignment mechanism were to consist of n_s independent assignments. For further discussion of this form of "conditional as-if analysis," see Pashley et al. (2021) and Rosenbaum (2017, pp. 289–290, fn. 15); Pashley et al. (2021, Section 5) also discuss why forming matched sets from the realized assignment complicates conditional as-if analysis (see Section 3.1).

We denote by Ω_s the set of all possible treatment assignments in set s, holding fixed the observed number of treated units in that set. Formally, Ω_s includes every possible way the n_s units in set s could be assigned to treatment and control such that exactly m_s units are treated. The number of possible assignments in Ω_s is denoted by $|\Omega_s|$, where the notation $|\cdot|$ indicates the number of elements in a set. This quantity equals $|\Omega_s| = \binom{n_s}{m_s} = n_s!/[m_s!(n_s - m_s)!]$, where "!" denotes the factorial operator (e.g., $4! = 4 \times 3 \times 2 \times 1$).

In the `ssafrica.3` set, for example, there are 4 observations — 1 treated and 3 control — so $|\Omega_s|$ for the `ssafrica.3` set is $\binom{4}{1} = 4$. The corresponding set of possible assignments with this treated count is shown in Table 2.

	Assignment 1	Assignment 2	Assignment 3	Assignment 4
Burundi	0	0	0	1
Rwanda	0	0	1	0
Somalia	0	1	0	0
Lesotho	1	0	0	0

Table 2: All possible treatment assignments within matched set `ssafrica.3`, holding fixed the observed number of treated units in that set.

The column labeled Assignment 3 shows the assignment that actually occurred. The other possible assignments — Assignment 1, Assignment 2, and Assignment 4 — represent cases in which Lesotho, Somalia, or Burundi is treated instead of Rwanda. The set excludes any assignments with more than 1 treated unit.

The set of possible treatment assignments across all matched sets, given the number treated in each, is $\Omega = \Omega_1 \times \cdots \times \Omega_S$, which consists of all combinations formed by choosing one assignment from each Ω_s.

Although the assignment itself can vary, the underlying causal quantities of interest — $\boldsymbol{\tau}$ and τ — remain fixed across all possible assignments. What changes from one assignment to another is which potential outcomes we would actually observe. In the observed Assignment 3, we see Rwanda's treated potential outcome and the control potential outcomes of Burundi, Somalia, and Lesotho, but not Rwanda's control potential outcome or the treated potential outcomes of the others. Under a different assignment, a different set of treated and control potential outcomes would have been observed. No matter which assignment occurs, we observe only partial information about our causal targets.

Inference from the partial information contained in the data to our causal targets is predicated on a probability distribution defined over the set of assignments, Ω. This distribution constitutes the uncertainty underlying our inferences — what Fisher (1935, p. 14) famously called the "reasoned basis" for inference — but, unlike in a randomized experiment, this distribution is unknown in an observational study. Because this distribution is unknown, we must make assumptions about it when drawing inferences from observational data. Under the assumption of as-if randomization, every possible assignment within each Ω_s is equally likely, making each overall assignment in Ω equally likely as well. In this case, all individuals within the same matched set have the same probability of treatment, equal to m_s/n_s. When as-if randomization does not hold, however, assignments are no longer equally likely, and some individuals have a higher probability of treatment than others.

2.3.1 How Do I Draw Inferences under the Sharp Framework?

To set the stage for inference under the sharp framework, consider a thought experiment. Suppose we were to subtract the true individual effects from the outcomes of whichever units happen to be treated. Doing so would yield, for each unit, the outcome it would have had under control. In other words, we may imagine reconstructing the dataset so that, under any possible treatment assignment, the outcomes would appear exactly as they would have if no one had been treated. In this reconstructed world, there would be no effect since every unit's outcome would reflect what it would have been without treatment.

Of course, we do not know the true collection of individual effects, $\boldsymbol{\tau}$, but we can test hypotheses about it, such as the hypothesis that all units share a common effect, whose hypothesized value we denote by τ_h. We do so by evaluating whether the data would look consistent with no treatment effect when that hypothesized value is subtracted from the treated outcomes. If, after this reconstruction, the data still show a positive effect, then the hypothesized value is presumably too small; if they show a negative effect, then the hypothesized value is presumably too large.

From a hypothesis testing perspective, when outcomes are reconstructed under a null hypothesis, the observed test statistic — computed on these reconstructed outcomes — will tend to fall in the upper tail of the null distribution if the hypothesized effect is too small, and in the lower tail if it is too large. To generate this null distribution, we reconstruct the outcomes under the hypothesized effect. Under the null, these reconstructed outcomes would remain fixed across assignments, so we hold them constant and recalculate the test statistic for every possible assignment. We then use this null distribution to determine where the observed test statistic falls for the calculation of p-values.

A canonical choice of test statistic in this setting is a *sum statistic*, which first adds up the treated outcomes within each set, and then adds those set-level sums across all sets. Many familiar test statistics can be expressed in this form by applying to the outcomes scale and shift transformations that do not depend on the treatment assignments. One such useful test statistic that can be expressed as a sum statistic is the Difference-in-Means, computed within sets and then averaged across sets, with each set weighted by its contribution to the effective sample size.

Under as-if randomization, the harmonic-mean-weighted Difference-in-Means is useful because, as discussed earlier, the variance of the within-set Difference-in-Means is minimized — under homogeneous treatment effects — when the harmonic mean of the numbers of treated and control units is largest. As a result, under this model of homogeneous individual treatment effects with similar outcome variances across sets, a test has greater power when the test places more weight on sets with large harmonic means — that is, sets in which the Difference-in-Means is most informative. Weighting sets solely by their share of units, by contrast, can overweight large but highly lopsided sets and dilute information from more balanced comparisons. When effects are heterogeneous, however, the harmonic-mean weights also change the quantity being estimated. The harmonic-mean-weighted Difference-in-Means estimates a harmonic-mean-weighted average of set-level effects, the matched-set analogue of the overlap-weighted average effect of Li et al. (2018), rather than the ATE targeted in the weak framework of Section 2.3.2.

The harmonic-mean-weighted Difference-in-Means also has the practical advantage of coinciding with two other common approaches to analyzing matched data:

- First, the harmonic-mean-weighted Difference-in-Means equals the coefficient on the treatment indicator from a fixed-effects regression that includes matched set indicators (Hansen and Bowers, 2008, pp. 228–229), an approach commonly used in practice after matching.
- Second, the harmonic-mean-weighted Difference-in-Means coincides with the overlap-weighted Difference-in-Means (Li et al., 2018) when overlap weights are constructed using the treatment assignment probabilities implied by as-if randomization. Overlap weights weight each control unit by the unit's probability of treatment and each treated unit by 1 minus that probability, so that the weighted treated group and the weighted control group share a common covariate distribution that gives the most weight to units with intermediate assignment probabilities. Because the assignment probabilities implied by as-if randomization are constant within each matched set, the resulting overlap weights are constant within the treated units and within the control units of each set, and these overlap weights aggregate exactly to the harmonic-mean weights implied by the matched sets.

To implement a sum statistic equivalent to the harmonic-mean-weighted Difference-in-Means for hypothesis testing, we first reconstruct the outcomes that treated units would have exhibited under the null hypothesis $\tau_h = 0$.

```r
# Keep only rows assigned to a matched set (drop NA in fm)
data_matched <- filter(
  .data = data, !is.na(fm)
)

# Null hypothesis value
tau_h <- 0

# Reconstruct outcomes under sharp null (tau_h = 0)
data_matched <- data_matched |>
  mutate(
    ldur_tilde = ldur - tau_h * UN
  )
```

We now source a helper function from the companion GitHub repository; this function rescales the outcome variable so that the sum of the rescaled values among treated units equals the harmonic-mean-weighted Difference-in-Means. After sourcing the function, we apply it to the matched dataset to generate a new column containing the rescaled outcomes. We then use this rescaled outcome to compute the observed test statistic.

```r
# Load hm_stat_rescale() from GitHub repo
# base_url (defined earlier) points to repo URL
source(
  paste0(
    base_url, "/R/hm_stat_rescale.R"
  )
)

# Apply rescaling: adds ldur_tilde_hm_scaled and returns matched dataset
data_matched <- hm_stat_rescale(
  data = data_matched,
  outcome = ldur_tilde, # Set outcome to be rescaled within matched sets
  treat = UN,           # Set name of treatment indicator variable
  strata = fm           # Set name of matched strata (block) variable
)

# Observed HM-weighted diff-in-means statistic
obs_stat <- sum(
  data_matched$ldur_tilde_hm_scaled[data_matched$UN == 1]
)
```

We can verify in `R` that this observed sum statistic coincides with the harmonic-mean-weighted Difference-in-Means, the coefficient on the treatment indicator from a fixed-effects regression that includes matched set indicators, and the overlap-weighted Difference-in-Means constructed from the assignment probabilities implied by as-if randomization.

First, we show that the observed sum statistic equals the harmonic-mean-weighted average of the within-matched-set Differences-in-Means, where each matched set is weighted by its contribution to the effective sample size.

```r
# Harmonic-mean-weighted difference in means (weights computed within sets)
dim_hm <- data_matched |>
  group_by(
    fm
  ) |>
  summarize(
    n_treated = sum(UN == 1),
    n_control = sum(UN == 0),
    # Within-set Difference-in-Means
    dim_set   = mean(ldur_tilde[UN == 1]) - mean(ldur_tilde[UN == 0]),
    # Harmonic-mean weight (set's contribution to effective sample size)
    w_hm      = 2 * n_treated * n_control / (n_treated + n_control),
    .groups   = "drop" # Drop grouping after summarize
  ) |>
  summarize(
    # Harmonic-mean-weighted average of within-set Differences-in-Means
    dim_hm = sum(
      w_hm * dim_set) / sum(w_hm)
  ) |>
  pull( # Pull column out of data frame
    dim_hm
  )

# Approach coincides with the sum statistic
all.equal(
  dim_hm, obs_stat
)
```

```
[1] TRUE
```

Next, we verify that the same quantity is obtained as the coefficient on the treatment indicator from a fixed-effects regression that includes matched set indicators.

```
# FE coefficient on UN (matched-set FE)
fe_fit <- lm(
  formula = ldur_tilde ~ UN + fm,
  data = data_matched
)

# Drop name so comparison is purely numeric
treat_coef_fe_fit <- unname(
  coef(fe_fit)["UN"]
)

# Approach coincides with the sum statistic
all.equal(
  treat_coef_fe_fit, obs_stat
)
```

```
[1] TRUE
```

Finally, we confirm that the observed sum statistic also coincides with the overlap-weighted Difference-in-Means. The overlap weights are constructed from the treatment assignment probabilities implied by as-if randomization within matched sets, with treated units weighted by one minus their treatment probability and control units weighted by their treatment probability.

```
# Install "PSweight" package (only run if not already installed)
# install.packages("PSweight")
library(PSweight) # For overlap weights from Li et al. (2018)
```

```
# Treatment probabilities within matched sets: n_treated / n
ps_fit <- PSmethod(
  ps.formula = UN ~ factor(fm), # Reproduces n_treated / n in each set
  method     = "glm",           # Estimate using generalized linear model
  # Use as.data.frame() for PSmethod() compatibility
  data       = as.data.frame(data_matched),
  ncate      = 2                # Binary treatment (treated vs. control)
)

assign_prob <- ps_fit$e.h[, "1"]  # Column for treatment level "1"

# Overlap weights from treatment probabilities
# Controls: treatment probability; Treated: 1 - treatment probability
w_ow <- ifelse(
  test = data_matched$UN == 1,
  yes  = 1 - assign_prob,
  no   = assign_prob
)

# Overlap-weighted difference in means
dim_ow <- data_matched |>
  summarize( # Aggregate rows into summary values
    # Overlap-weighted mean outcome for treated units
    treated_weighted_mean =
      sum(w_ow[UN == 1] * ldur_tilde[UN == 1]) / sum(w_ow[UN == 1]),

    # Overlap-weighted mean outcome for control units
    control_weighted_mean =
      sum(w_ow[UN == 0] * ldur_tilde[UN == 0]) / sum(w_ow[UN == 0]),

    # Difference between overlap-weighted treated and control means
    dim_ow = treated_weighted_mean - control_weighted_mean
  ) |>
  pull(
    dim_ow
  )

# Approach coincides with the sum statistic
all.equal(
  dim_ow, obs_stat
)
```

```
[1] TRUE
```

Now, to generate the distribution to which we refer our observed sum statistic, we can hold the reconstructed and rescaled outcomes fixed and then calculate the sum statistic over all possible assignments in Ω. In our application, with 12 sets ranging in size from 2 to 5, the total number of assignments is 311,040.

```r
# For each matched set (fm), record:
# n = total units in the set
# m = number treated (UN == 1)
block_ns <- data_matched |>
  group_by(          # Group results by key variables
    fm
  ) |>
    summarize(         # Aggregate to one row per group
    n = n(),         # Row count per group
    m = sum(UN),
    .groups = "drop"
  )

# Total possible treatment assignments = product of binomial coefficients
# (choose m_s of the n_s units for treatment in each set and multiply across sets)
exact_n_assigns <- prod(
  choose(
    n = block_ns$n,
    k = block_ns$m
  )
)
```

Because our matched study is relatively small, we can enumerate all possible treatment assignments exactly.

```r
# Install "randomizr" (only run if not already installed)
# install.packages("randomizr")

# Load randomizr for generating random assignments
library(randomizr)

exact_assigns <- obtain_permutation_matrix(
  declaration = declare_ra(   # Declare assignment procedure
    N = nrow(data_matched),   # Total number of units
    blocks = data_matched$fm, # Matched set membership
    block_m = block_ns$m      # Number treated in each set
  ),
# Total number of assignments with treated counts fixed within sets
  maximum_permutations = exact_n_assigns
)
```

However, in most applications, exactly enumerating all possible assignments is computationally infeasible.

Instead, we typically draw a random subset of assignments (e.g., 10,000) to approximate the exact randomization distribution, as the following code illustrates.

```r
# Set RNG seed for reproducibility
set.seed(11242017)

# Generate permutation matrix of treatment assignments
# Each column = one possible assignment consistent with block structure
sim_assigns <- obtain_permutation_matrix(
  declaration = declare_ra(
    N = nrow(data_matched),
    blocks = data_matched$fm,
    block_m = block_ns$m
  ),
  maximum_permutations = 10^4  # Cap at 10,000 random draws
)
```

Now, to generate the null distribution of the sum statistic, we apply the statistic to each of these 10,000 assignments while holding the reconstructed and rescaled outcomes fixed under the null.

```r
# Randomization distribution under sharp null
# Apply sum statistic to each assignment in sim_assigns
# Outcome transformed so sum = harmonic-mean-weighted difference-in-means
sim_sharp_null_dist <- apply(
  X = sim_assigns,     # Matrix of treatment assignments
  MARGIN = 2,          # Iterate over columns (assignments)
  FUN = function(x) {
    # Sum transformed outcomes among treated
    sum(
      data_matched$ldur_tilde_hm_scaled[x == 1]
    )
  }
)
# Faster equivalent computation via matrix multiplication
# as.numeric(
#   t(data_matched$ldur_tilde_hm_scaled) %*% sim_assigns
# )
```

From this null distribution, we compute a one-sided upper p-value, which, under as-if randomization, is simply the proportion of null test statistics greater than or equal to the observed test statistic.

```r
# One-sided, upper p-value:
# Proportion of simulated randomization stats >= observed
round(
  x = mean(
    sim_sharp_null_dist >= obs_stat
  ),
  digits = 4
)
```

```
[1] 0.0346
```

The upper p-value of a test of the sharp null of no effect against the alternative of a larger effect is 0.0346. (Counting the observed assignment as one of the draws, $(1 + 346)/(1 + 10{,}000) = 0.0347$, gives a p-value that is valid by construction.)

In this particular case, unlike in most applications, the matched study is small enough to compute the exact p-value and assess the accuracy of the simulation-based p-value approximation.

```
# Transformed outcomes under sharp null (tau_h = 0)
q_tau_h_0 <- data_matched$ldur_tilde_hm_scaled

# Fast computation of exact null distribution using matrix multiplication
exact_sharp_null_dist <- as.numeric(
  t(q_tau_h_0) %*% exact_assigns
)

# Slower apply()-based computation (for reference)
# apply(
#  X = exact_assigns,
#  MARGIN = 2,
#  FUN = function(x) {
#    sum(data_matched$ldur_tilde_hm_scaled[x == 1])
#  }
#)

# Exact one-sided, upper p-value
round(
  x = mean(
    exact_sharp_null_dist >= obs_stat
  ),
  digits = 4
)
```

```
[1] 0.0363
```

The exact p-value of 0.0363 is nearly identical to the simulation-based p-value of 0.0346. At the conventional significance level of $\alpha = 0.05$, we would reject the null hypothesis in favor of the alternative, regardless of whether we use the exact or simulation-based p-value.

As an alternative to randomly sampling assignments and computing the test statistic for each one, we can use a much faster Normal approximation to the null distribution when the matched study contains many matched sets. The approximation relies on closed-form expressions for the expected value and variance of a sum statistic derived in Rosenbaum and Krieger (1990). Using these expressions, we standardize the observed test statistic and then compare it to the standard Normal distribution, which gives us a corresponding p-value. Although primarily designed for sensitivity analyses under violations of as-if randomization, the `senstrat` package can also be used to compute the null expected value and variance of a sum statistic under as-if randomization.

```
# Install "senstrat" package (only run if not already installed)
# install.packages("senstrat")

# Load senstrat for stratum-level null moments and sensitivity analysis
# (Rosenbaum & Krieger 1990)
library(senstrat)

# Compute per-block null expectations and variances
per_block_moms <- data_matched |>
  group_by(
    fm
  ) |>
  summarize(
    expect   = ev(
      sc = ldur_tilde_hm_scaled, # Transformed outcomes for the stratum
      z = UN,                     # Treatment indicator
      m = 1,                      # Count of 1s in hidden confounder
                                  # Irrelevant here since Gamma = 1

      g = 1,                      # Sensitivity parameter Gamma
      method = "RK"               # Use Rosenbaum and Krieger (1990) formula
    )$expect,                     # Null expectation of sum statistic by set
    variance = ev(
      sc     = ldur_tilde_hm_scaled,
      z      = UN,
      m = 1,
      g      = 1,
      method = "RK"
    )$vari,                       # Null variance of sum statistic by set
    .groups  = "drop"
  )

# Sum across blocks to get overall null expectation
null_ev  <- sum(
  per_block_moms$expect
)

# Sum across blocks to get overall null variance
null_var <- sum(
  per_block_moms$variance
)

# Standardized test statistic and one-sided Normal p-value (upper tail)
norm_upper_p_value <- pnorm(
  q = (obs_stat - null_ev) / sqrt(null_var), # Standardized statistic
  lower.tail = FALSE                         # Compute upper-tail prob
)
# By default in pnorm(): mean = 0 and sd = 1 (standard normal distribution)

# Display the Normal-approximation p-value
round(
  x = norm_upper_p_value,
  digits = 4
)
```

```
[1] 0.0343
```

This Normal-approximation p-value of 0.0343 is close to the exact and simulation-based p-values of 0.0363 and 0.0346, respectively. All three p-values lead us to reject the sharp null of no effect in favor of a larger effect.

To test not only a single null hypothesis but a grid of null hypotheses under the sharp framework, we repeat the Normal-approximation test for a range of hypothesized constant effects, τ_h. For each value of τ_h, we reconstruct the outcomes under the null, rescale them so that the sum statistic equals the harmonic-mean-weighted Difference-in-Means, compute the observed test statistic and the statistic's null expectation and variance, and obtain the corresponding p-value using the Normal approximation. To form a one-sided confidence set, we retain all null values that are not rejected by the upper-tail test at the chosen α level. To form a two-sided confidence set, we proceed analogously, allocating $\alpha/2$ to each tail — reflecting that each null value is tested against departures in both directions — and using the corresponding upper- and lower-tail rejection regions.

```
# Significance level
alpha <- 0.05

# Grid of constant-effect null values for the Normal approximation
tau_h_grid_norm <- seq(
  from = -0.5,
  to = 2,
  by = 0.001
)

# Repeat the Normal-approximation test for each tau_h
# sapply() computes and stacks the resulting p-values
p_mat_norm <- sapply(X = tau_h_grid_norm,
                     FUN = function(tau_h) {

                       # Reconstruct outcomes under null tau_h and rescale them
                       dat_tau_h <- hm_stat_rescale(
                         data     = transform(
                           data_matched,
                           ldur_tilde_shift = ldur - tau_h * UN),
                         outcome = ldur_tilde_shift,
                         treat   = UN,
                         strata  = fm
                       )

                       # Observed statistic under null tau_h
                       obs_stat_tau_h <- sum(
                         dat_tau_h$ldur_tilde_shift_hm_scaled[dat_tau_h$UN == 1]
                       )

                       # Null expectation and variance of the statistic under tau_h
                       moms_tau_h <- dat_tau_h |>
                         group_by(
                           fm
                         ) |>
                         summarize(
                           expect   = ev(sc = ldur_tilde_shift_hm_scaled, z = UN,
                                         m = 1, g = 1, method = "RK")$expect,
                           variance = ev(sc = ldur_tilde_shift_hm_scaled, z = UN,
                                         m = 1, g = 1, method = "RK")$vari,
                           .groups  = "drop"
                         )

                       # Standardized statistic under null tau_h
                       std_stat_tau_h <- (obs_stat_tau_h - sum(moms_tau_h$expect)) /
                         sqrt(sum(moms_tau_h$variance))

                       # Upper-tail and lower-tail Normal-approximation p-values
                       c(
                         upper_tail = pnorm(q = std_stat_tau_h, lower.tail = FALSE),
                         lower_tail = pnorm(q = std_stat_tau_h)
                       )
                     })
```

```
# Upper-tail confidence set: retain all tau_h with upper-tail p-value >= alpha
cs_sharp_upper_tail_norm <- c(
  lower = min(
    tau_h_grid_norm[p_mat_norm["upper_tail", ] >= alpha]
  ),
  upper = Inf
)

# Two-sided confidence set: retain tau_h only if neither one-sided test
# rejects at level alpha / 2
cs_sharp_two_sided_norm_vals <- tau_h_grid_norm[
  p_mat_norm["upper_tail", ] >= alpha / 2 & p_mat_norm["lower_tail", ] >= alpha / 2
]

# Two-sided confidence set summarized by its bounds
cs_sharp_two_sided_norm <- c(
  lower = min(
    cs_sharp_two_sided_norm_vals
  ),
  upper = max(
    cs_sharp_two_sided_norm_vals
  )
)
```

An analogous procedure applies when constructing confidence sets without relying on a Normal approximation, instead using a simulation-based approximation to the null randomization distribution.

```
# Grid of constant-effect null values (sharp framework)
tau_h_grid <- seq(
    from = -0.5,
  to = 1.5,
  by = 0.0001
)

# Repeat the randomization test for each tau_h
# sapply() computes and stacks the resulting p-values
p_mat <- sapply(X = tau_h_grid,
                FUN = function(tau_h) {

                  # Shift outcomes under null: ldur_i - tau_h * UN_i
                  dat_tau_h <- hm_stat_rescale(
                  # transform(): copy data_matched adding shifted outcome
                    data    = transform(
                      data_matched,
                      ldur_tilde_shift = ldur - tau_h * UN),
                    outcome = ldur_tilde_shift,
```

```r
                    treat    = UN,
                    strata   = fm
                  )

                  # Transformed outcomes under sharp null tau_h
                  q_tau_h <- dat_tau_h$ldur_tilde_shift_hm_scaled

                  # Observed statistic under null tau_h
                  obs_stat_tau_h <- sum(
                    dat_tau_h$UN * q_tau_h
                  )

                  # Randomization distribution via simulated assignments
                  # Matrix multiplication used for speed (vs apply())
                  sim_null_dist_tau_h <- as.numeric(
                    t(q_tau_h) %*% sim_assigns
                  )

                  # Upper-tail and lower-tail randomization p-values
                  p_upper_tau_h <- mean(
                    sim_null_dist_tau_h >= obs_stat_tau_h
                  )

                  p_lower_tau_h <- mean(
                    sim_null_dist_tau_h <= obs_stat_tau_h
                  )

                  c(
                    upper_tail = p_upper_tau_h,
                    lower_tail = p_lower_tau_h
                  )
                })

# Extract vector of upper-tail p-values
p_upper <- p_mat["upper_tail", ]

# Extract vector of lower-tail p-values
p_lower <- p_mat["lower_tail", ]

# Simulation-based confidence sets

# Upper-tail confidence set: retain all tau_h with upper-tail p >= alpha
cs_sharp_upper_tail_sim <- tau_h_grid[p_upper >= alpha]

# Lower bound of the upper-tail confidence set
cs_sharp_upper_tail_sim_bound <- min(
  cs_sharp_upper_tail_sim
)
```

```
# Two-sided confidence set (inversion using alpha/2 in each tail):
# retain tau_h only if neither one-sided test rejects at level alpha/2
cs_sharp_two_sided_sim <- tau_h_grid[
  p_upper >= alpha / 2 & p_lower >= alpha / 2
]

# Two-sided confidence set summarized by its bounds
cs_sharp_two_sided_sim_bounds <- c(
  lower = min(
    cs_sharp_two_sided_sim
  ),
  upper = max(
    cs_sharp_two_sided_sim
  )
)
```

With these simulated assignments, the one-sided 95% confidence set for a constant effect is $[0.069, \infty)$ and the two-sided 95% confidence set is $[-0.066, 1.358]$. The Normal approximation gives $[0.077, \infty)$ and $[-0.064, 1.364]$, respectively.

Likewise, for a point estimate, one could follow Hodges and Lehmann (1963) and Rosenbaum (1993) by identifying the null value that makes the observed sum statistic equal to its null expectation. With our test statistic under as-if randomization, the null expectation is 0 for every τ_h, and the observed statistic equals 0 exactly when τ_h equals the harmonic-mean-weighted Difference-in-Means computed from the observed outcomes (roughly 0.673). This harmonic-mean-weighted Difference-in-Means is therefore the Hodges–Lehmann estimate.

2.3.2 How Do I Draw Inferences under the Weak Framework?

When the target is the ATE, it can be written as $\tau = \sum_{s=1}^{S} (n_s/n)\tau_s$, a weighted average of the set-level ATEs, $\tau_s = (1/n_s) \sum_{i=1}^{n_s} \tau_{si}$, with weights equal to each set's share of the total study size. Thus, a straightforward way to estimate the ATE is to compute the Difference-in-Means within each matched set, and then combine these set-level estimates using the same weights. Formally, the overall Difference-in-Means is $\hat{\tau} = \sum_{s=1}^{S} (n_s/n)\hat{\tau}_s$, where $\hat{\tau}_s$ is the Difference-in-Means in set s.

These weights are appealing because, under as-if randomization, the resulting Difference-in-Means is an unbiased estimator of the ATE. Formally, the expected value of the estimator — i.e., the average of $\hat{\tau}$

taken over all treatment assignments consistent with the matched design and their associated probabilities — equals the true ATE. Nevertheless, although correct in expectation, the Difference-in-Means can vary substantially across assignments. In any given assignment, the estimate may lie far from the target ATE.

Because the estimator can fluctuate across different treatment assignments, it is important to quantify the typical squared distance between an estimate and the true ATE. To this end, Neyman (1923) introduced a canonical variance estimator. Under as-if randomization, this estimator is exactly unbiased when individual treatment effects are homogeneous; otherwise, it is conservative, meaning its expected value is greater than or equal to the true variance of the Difference-in-Means, although subsequent refinements reduce this conservativeness while maintaining validity (see Robins, 1988; Aronow et al., 2014; Harshaw et al., 2024). In principle, we could apply this approach by estimating the variance of the Difference-in-Means within each matched set and then taking a weighted sum of these estimates (see, e.g., Miratrix et al., 2013).

In our setting, however, this approach is infeasible because each matched set contains only 1 treated or 1 control unit. The usual Neyman formula relies on computing sample variances separately within the treated and control groups of each set. The sample variance requires at least two observations because `var()` divides by the number of observations minus 1. As a result, we cannot compute the sample variance when there is only a single treated or control unit.

How, then, can we estimate the variance of the Difference-in-Means in a matched observational study? Pashley and Miratrix (2021) and Fogarty (2018) offer distinct solutions. Pashley and Miratrix (2021) propose two approaches, each valid under different conditions on the matched structure, while Fogarty (2018) develops a single method that applies more broadly to finely stratified designs.

The first approach in Pashley and Miratrix (2021), `hybrid_m`, requires at least two matched sets of each unique set size in the data. The second approach, `hybrid_p`, permits variation in set sizes as long as no single set contains half or more of the units in sets with only one treated or one control unit (here, all units). Our matched design meets the condition for the second approach, not the first. We therefore estimate the variance using `hybrid_p` through the `blkvar` package, the companion software for Pashley and Miratrix (2021).

```r
# Install blkvar (only run if not already installed)
# install.packages("remotes")
# remotes::install_github("lmiratrix/blkvar")

# Load blkvar for block randomization variance estimators
library(blkvar)

# Compute results with hybrid_p method
res <- block_estimator(
  Yobs = ldur,          # Observed outcomes
  Z = UN,               # Treatment indicator
  B = fm,               # Block (matched set) membership
  data = data_matched, # Dataset
  method = "hybrid_p"  # Variance estimation method
)

# Extract variance estimate
res$var_est
```

```
[1] 0.1187282
```

We can also estimate the variance of the Difference-in-Means in a finely stratified design using the approach of Fogarty (2018), using each set's size in the estimator's projection step, as in Fogarty (2023, p. 2199). To do so, we load a custom R function, `fine_strat_var_est()`, from the companion GitHub repository.

```r
# Load the fine_strat_var_est() function from the GitHub repo
source(
  paste0(
    base_url, "/R/fine_strat_var_est.R"
  )
)
```

The function takes as inputs the set-specific treatment effect estimates and the number of units in each set, and it returns a single scalar representing the estimated variance. We compute the set-specific estimates and corresponding set sizes, and then pass these two vectors as inputs to `fine_strat_var_est()`.

```r
# Compute matched set sizes and matched-set-specific differences in means
set_stats <- data_matched |>
  group_by(
    fm
  ) |>
  summarize(
    n = n(),                                     # Matched set size
    diff_in_means = mean(                        # Treated mean
      ldur[UN == 1L]
    ) -                                          # minus
      mean(                                      # control mean
        ldur[UN == 0L]
      ),
    .groups = "drop"
  )

# Apply Fogarty (2018/2023) variance estimator
fine_strat_var_est(
  strat_ns = set_stats$n,                      # Vector of stratum sizes
  strat_ests = set_stats$diff_in_means         # Vector of stratum estimates
)
```

```
[1] 0.1163934
```

This variance estimate is nearly identical to the one we obtained using the `hybrid_p` approach of Pashley and Miratrix (2021).

Now that we have estimates of both the ATE and the variance of the ATE estimator, we can form a standardized test statistic by subtracting the expected Difference-in-Means implied by the null and dividing the result by the estimated standard error (the square root of the estimated variance). We then compare this statistic to a standard Normal distribution to calculate p-values. The following code calculates the upper one-sided p-value for a test of the null hypothesis that the ATE is 0 against the alternative that the ATE is positive.

```
pnorm(
  q = (res$ATE_hat - 0) / sqrt(res$var_est),
  lower.tail = FALSE
)
```

```
[1] 0.03722157
```

Here, we reject the weak null hypothesis that the ATE equals 0 in favor of the alternative that the ATE is greater than 0, using the conventional significance level of $\alpha = 0.05$.

As in the construction of confidence sets for a homogeneous treatment effect under the sharp framework, we form 95% confidence sets by inverting the corresponding hypothesis tests. We first construct a one-sided confidence set obtained by inverting the upper-tail test and retaining all null values that are not rejected at the $\alpha = 0.05$ level. We then construct a two-sided confidence set by allocating $\alpha/2 = 0.025$ to each tail, as in the tail allocation used in the sharp framework.

```r
# One-sided (upper-tail) confidence set
cs_weak_upper_tail <- c(
  lower = res$ATE_hat - qnorm(p = 1 - alpha) * sqrt(res$var_est),
  upper = Inf
)

# Two-sided confidence set
cs_weak_two_sided <- c(
  lower = res$ATE_hat - qnorm(p = 1 - alpha / 2) * sqrt(res$var_est),
  upper = res$ATE_hat + qnorm(p = 1 - alpha / 2) * sqrt(res$var_est)
)
```

The estimated ATE (`res$ATE_hat`) is 0.615, with an estimated standard error of 0.345. The resulting one-sided 95% confidence set is $[0.048, \infty)$. The two-sided 95% confidence set, $[-0.061, 1.290]$, includes 0 because the one-sided p-value of 0.037 exceeds $\alpha/2 = 0.025$.

2.4 Third Stage: Sensitivity Analysis (Departures from As-If Randomization)

Up to this point, we have assumed a framework in which each observation's chance of a UN intervention is like flipping a weighted coin. Each coin flip is independent across observations, but the probability of landing tails (i.e., receiving treatment) can differ from one unit to another depending on its baseline characteristics. With matching, we make our crucial assumption that all units within a set are similar enough on these characteristics that their coins have the same probability of landing tails.

This as-if randomization assumption may fail because of imbalances on hidden covariates (or residual imbalances on observed covariates, though the sensitivity analysis to follow subsumes both within the same framework). To represent such imbalances, consider a single hidden covariate, $\boldsymbol{u} = (u_{11}, \dots, u_{Sn_S})^\top$, indexed first by matched set and then by unit within set, with each u_{si} constrained to lie in the interval from 0 to 1. Although the restriction that each element of $\boldsymbol{u}$ is between 0 and 1 may seem strong, Rosenbaum (2017, p. 300, fn. 33) shows that any departure from complete random assignment within blocks can be represented by such a $\boldsymbol{u}$ in the sense that as-if randomization would hold if we had access to and exactly matched on this $\boldsymbol{u}$.

Rosenbaum (1987a) and Rosenbaum and Krieger (1990) then propose a model in which each unit's independent probability of treatment assignment is given by

$$\pi_{si} = \frac{\exp\left[\kappa_s + \log(\Gamma) u_{si}\right]}{1 + \exp\left[\kappa_s + \log(\Gamma) u_{si}\right]}. \tag{3}$$

The parameter κ_s is a set-specific intercept that captures the baseline propensity for treatment shared by all units in matched set s before accounting for any differences in the hidden covariate. In other words, κ_s reflects the central idea in matching that, after forming matched sets homogeneous in observed covariates, all individuals within a set share the same treatment propensity based on those covariates. The parameter $\Gamma \geq 1$, by contrast, quantifies how strongly the hidden covariate u_{si} can alter treatment odds. When $\Gamma = 1$, all individuals in matched set s have the same probability of treatment, $\exp[\kappa_s]/(1 + \exp[\kappa_s])$. However, when $\Gamma > 1$, two individuals in the same set who differ in the hidden covariate may differ in their odds of treatment by as much as a factor of Γ.

An important point to reiterate is that we condition on assignments that belong to the set Ω, meaning that the number of treated units is fixed within each matched set. This conditioning removes dependence on the set-specific baseline κ_s in the probability distribution over assignments in Ω (Rosenbaum, 1984; Rosenbaum and Krieger, 1990). Eliminating this dependence is crucial because, without the set-specific intercepts, we can characterize both as-if randomization and departures from as-if randomization using only a single sensitivity parameter, Γ.

To build intuition for the sensitivity parameter Γ, note that the model in (3) implies the following restriction:

$$\frac{1}{\Gamma} \leq \frac{\pi_{si}\left(1 - \pi_{sj}\right)}{\pi_{sj}\left(1 - \pi_{si}\right)} \leq \Gamma \text{ for all } i,\ j \text{ and } s. \tag{4}$$

This restriction states that, within any set, no two units can differ in their odds of treatment by more than a factor of Γ. When $\Gamma = 1$, all units share the same odds of treatment, corresponding to as-if randomization. By contrast, larger values of Γ represent increasingly severe departures from this assumption.

Rosenbaum (1995, pp. 1424–1425) shows that the converse also holds, that is, that the restriction in (4) implies a model of the form in (3). For any collection of treatment probabilities satisfying the bound in (4), it is always possible to find values of $u_{si} \in [0, 1]$ for all sets s and units i, together with set-specific intercepts κ_s, such that (3) holds with the same Γ and therefore yields the same probability distribution over assignments in Ω. In other words, (4) describes a general restriction on treatment probabilities that does not assume a particular functional form. The logistic model in (3), by contrast, provides one convenient parametric representation of that general restriction.

Thus far, we have conducted inference assuming $\Gamma = 1$, meaning that all units within a matched set share the same treatment probability. Under this as-if randomization assumption, tests of sharp null hypotheses are exactly valid, while tests of weak null hypotheses are valid asymptotically, as the number of matched sets grows with set sizes bounded (Fogarty, 2018, 2023). In both cases, validity means that the probability of falsely rejecting the null hypothesis does not exceed the nominal level α.

When we allow departures from as-if randomization, governed by the sensitivity parameter $\Gamma \geq 1$, we seek new p-values that remain valid in the same sense. The probability of a false rejection should not exceed α, regardless of the hidden covariate $\boldsymbol{u}$ or the potential outcomes consistent with the null. The way we ensure this validity, however, differs between the sharp and weak frameworks, which we consider in turn.

2.4.1 How Do Inferences under the Sharp Framework Change under These Departures?

For a sharp null, if the hidden $\boldsymbol{u}$ were known, we could calculate from (3) the exact distribution of assignments consistent with the matched design for any given value of $\Gamma \geq 1$. This distribution would provide the correct reference for computing p-values under the null. In particular, the upper one-sided p-value could be obtained by summing the probabilities of all assignments whose test statistic under the null is at least as large as the observed statistic.

In practice, however, $\boldsymbol{u}$ is unknown. Thus, to ensure validity of our tests, we compute p-values under the worst-case choice of $\boldsymbol{u}$ — the one that makes the p-value as large as possible. Rejection under the worst-case choice of $\boldsymbol{u}$ guarantees rejection under the actual (but hidden) $\boldsymbol{u}$. Consequently, we will not reject the null more often than we would if the true assignment probabilities — governed by the actual but unobserved $\boldsymbol{u}$ — were known under the model in (3).

2.4.1.1 Finding the Worst-Case Scenario of Hidden Confounding for Valid Inference

Finding this worst-case $\boldsymbol{u}$ exactly is difficult. As a first step, Rosenbaum and Krieger (1990) show that in the unmatched (i.e., two-sample) case, for a broad class of test statistics that includes sum statistics, the vector $\boldsymbol{u}$ that maximizes the p-value under any fixed $\Gamma \geq 1$ must belong to a restricted set of possibilities, denoted by $\mathcal{U}_+$. Once the subjects are ordered from the largest to the smallest outcome, all the candidate vectors in $\mathcal{U}_+$ look the same: a sequence of 1s at the top followed by 0s below. We do not know *how many* 1s should precede the 0s, and this number determines which configuration of $\boldsymbol{u}$ yields the worst-case p-value. In an unmatched study, it is straightforward to enumerate all $n - 1$ such candidate vectors in $\mathcal{U}_+$, and then identify which one yields the largest p-value for a fixed $\Gamma \geq 1$.

Unfortunately, in matched designs, the overall worst-case vector $\boldsymbol{u}$ cannot be obtained by simply stitching together the worst-case vectors from each matched set. That is, the global worst case is not simply the collection of local worst cases. Instead, $\mathcal{U}_+$ consists of $\prod_{s=1}^{S}(n_s - 1)$ total candidate vectors for the worst-case $\boldsymbol{u}$ — a quantity that quickly becomes infeasible to enumerate directly. For example, with only 20 matched sets of 4 units each, the number of candidate vectors is already nearly 3.5 billion ($3^{20} = 3{,}486{,}784{,}401$).

2.4.1.1.1 Separable Approximation To address this challenge, Gastwirth et al. (2000) propose a practical shortcut called the *separable approximation.* The idea is to select, within each matched set, the configuration $\boldsymbol{u}_s$ (from the set-specific candidate sequences of 1s and 0s) that maximizes the expected value of the test statistic in that set under the null. If more than one candidate yields the same expectation, the choice goes to the $\boldsymbol{u}_s$ that produces the larger variance of the test statistic in that set under the null. The method is called "separable" because it then stitches together the choices of $\boldsymbol{u}_s$ made separately within each matched set, rather than searching over all possible combinations across sets. The resulting $\boldsymbol{u}$ may not give the exact worst-case p-value; yet as the number of matched sets grows, with set sizes bounded, the error becomes negligible (Gastwirth et al., 2000, Proposition 1).

2.4.1.1.2 Taylor Series Approximation The separable approximation is a useful shortcut, but the separable approximation's p-value is never larger than, and typically slightly smaller than, the worst-case p-value over $\mathcal{U}_+$ computed with the same Normal approximation (Rosenbaum, 2018, p. 2321). No result guarantees that the difference between these two p-values is negligible in designs with relatively few matched sets or with highly unbalanced set sizes. Rosenbaum (2018) therefore introduces a refinement that is never less conservative than this worst case, so that the resulting test is valid whenever the Normal approximation is adequate.

The key idea is to reframe hypothesis testing in terms of a function that, for each configuration of hidden confounding $\boldsymbol{u} \in \mathcal{U}_+$, combines the gap between the null expectation and the observed test statistic with a corresponding Normal critical buffer. Concretely, for $\alpha = 0.05$ and corresponding critical value 1.64, this function is the expected value of the test statistic under $(\boldsymbol{u}, \Gamma)$ minus the observed test statistic, plus 1.64 times the standard deviation of the test statistic under $(\boldsymbol{u}, \Gamma)$. The function is nonpositive exactly when the observed test statistic lies at least 1.64 standard deviations above its null expectation, and positive otherwise. Therefore, for a given $(\boldsymbol{u}, \Gamma)$, the Normal-approximation test rejects the null hypothesis if and only if this function is nonpositive, and the sensitivity analysis at Γ rejects if and only if the function is nonpositive for every $\boldsymbol{u} \in \mathcal{U}_+$, that is, if and only if the function's maximum over $\mathcal{U}_+$ is nonpositive.

We can code this function in `R` as follows.

```r
# Rosenbaum (2018) objective: expectation - observed + kappa * SD
# Nonpositive values imply rejection at level alpha
sens_objective <- function(null_expect,
                           null_variance,
                           obs_stat,
                           alpha = 0.05) {
  # null_expect   : overall null expectation of the test statistic
  # null_variance : overall null variance of the test statistic
  # obs_stat      : observed value of the test statistic
  # alpha         : test size (e.g., 0.05 gives kappa about 1.64)

    # Normal critical value (not the set intercept kappa_s in (3))
  kappa  <- qnorm(
    p = 1 - alpha
  )

  # Null standard deviation
  null_sd <- sqrt(
    null_variance
  )

  # Gap between null expectation and observed statistic
    # (null_expect - obs_stat) + Normal buffer (kappa * null_sd)
  obj_val <- (null_expect - obs_stat) + kappa * null_sd

  return(obj_val)
}
```

To build intuition, we next evaluate this function at $\Gamma = 1$.

```
# Evaluate under Gamma = 1 using null_ev, null_var, and obs_stat
sens_objective(
  null_expect   = null_ev,
  null_variance = null_var,
  obs_stat      = obs_stat,
  alpha         = 0.05
)
```

```
[1] -0.06500212
```

The resulting value is negative, indicating rejection of the null at the chosen significance level. This is exactly what we would expect. Because we previously rejected the sharp null hypothesis that $\tau_h = 0$ for all units at $\Gamma = 1$, the function evaluated at $\Gamma = 1$ is indeed nonpositive.

The crucial property of this function is concavity. Although $\mathcal{U}_+$ is a finite set, the function extends to a concave function of continuous weights placed on the candidate configurations within each set, because the null expectation and variance are linear in these weights and the only nonlinear term is the square root of the variance (Rosenbaum, 2018, p. 2323). Hence a tangent drawn at *any* $\boldsymbol{u} \in \mathcal{U}_+$ lies above the function at *every* configuration in $\mathcal{U}_+$. Rosenbaum (2018) draws the tangent line at the configuration of $\boldsymbol{u}$ chosen by the separable approximation. Because the original function is concave, this tangent line provides a global upper bound on the function for all candidate configurations of hidden confounding. If this upper bound is nonpositive, then the function itself must also be nonpositive. Consequently, rather than maximizing a curved function over $\mathcal{U}_+$, we can instead maximize a linear function, which greatly simplifies the optimization.

The tangent is easy to work with because it is linear in the set-specific null expectations and variances, so it decomposes additively across matched sets. We can therefore determine, within each matched set separately, which hidden-bias pattern makes the tangent line as large as possible, and then add up these set-specific contributions. The resulting sum is the maximum value of the tangent line over all $\boldsymbol{u} \in \mathcal{U}_+$.

Recalling that the original function is concave, if the maximum value of the tangent-line approximation over $\boldsymbol{u} \in \mathcal{U}_+$ is at or below zero, then the original function itself must be nonpositive for every $\boldsymbol{u} \in \mathcal{U}_+$. Therefore, within the Normal approximation, we can safely base the decision to reject the null hypothesis or not on the Taylor approximation. Although valid, this procedure can be conservative, because the tangent line may remain above zero even when the original function would be below zero at the true worst-case $\boldsymbol{u}$. By contrast, the separable approximation can be liberal, as it may reject even when the worst-case value of the function is positive.

2.4.1.2 Conducting Worst-Case Sensitivity Analysis We next report the upper one-sided p-value obtained under the separable approximation, along with an indication of whether the resulting decision to reject or not reject the null hypothesis agrees with the decision implied by the Taylor series approximation. Agreement or disagreement between the two methods is determined by the sign of the Taylor-series-based bound on the objective function (computed by `sens_objective()`) that governs rejection of the null hypothesis. Disagreement can occur only when the separable approximation yields a p-value below α (in practice, typically only slightly below), indicating rejection, while the Taylor series approximation yields a positive value, leading to non-rejection.

```r
# Grid of Gamma values
Gamma_vals <- seq(
  from = 1,
  to = 1.5,
  by = 0.0001
)

# lapply() runs sensitivity analysis at each Gamma, returns list of results
sens_results <- lapply(
  X = Gamma_vals,
  FUN = function(g) {
    out <- senstrat(
      sc          = data_matched$ldur_tilde_hm_scaled, # outcome
      z           = data_matched$UN,                   # treatment indicator
      st          = data_matched$fm,                   # matched set
      gamma       = g,                                 # Gamma
            alternative = "greater",                         # upper-tail test
      level       = alpha,                             # test level for cvA
      detail      = TRUE # return intermediate quantities
    )

    # Separable p-value
    p_sep <- as.numeric(
      out$Separable["P-value"]
    )

    # Taylor series margin (cvA)
    cvA <- as.numeric(
      out$lambda["Linear Taylor bound"]
    )

    # Decisions
    sep_reject <- p_sep <= alpha          # separable rejects?
    tay_reject <- cvA <= 0                # Taylor rejects?

    # Store results for this value of Gamma
    data.frame(
      Gamma     = g,        # sensitivity parameter
      p_sep     = p_sep,    # separable p-value
      cvA       = cvA,      # Taylor bound
      agree_tay = (sep_reject == tay_reject) # agreement indicator
    )
  }
)

# Bind into a single data frame
sens_df <- do.call(
  what = rbind,
  args = sens_results
)

# Smallest Gamma where the separable p-value is >= alpha
sens_value_sep <- min(
  sens_df$Gamma[sens_df$p_sep >= alpha]
)

# Taylor series sensitivity value:
# smallest Gamma at which Taylor margin becomes positive (non-rejection)
sens_value_tay <- min(
  sens_df$Gamma[sens_df$cvA > 0]
)
```

For this matched dataset, the conclusions from the separable and Taylor series approximations agree for nearly all values of Γ, differing over only a narrow interval from 1.1361 to 1.1399. As these results indicate, we can reject the sharp null of no effect under as-if randomization (i.e., $\Gamma = 1$), but this conclusion is sensitive to departures from that assumption. Using the Normal approximation together with the more conservative Taylor series approximation, we can no longer reject the sharp null of no effect against the alternative of a larger effect at the $\alpha = 0.05$ level for values of Γ of approximately 1.14 or larger.

Figure 3 illustrates this comparison and clarifies how the two approaches relate across values of Γ.

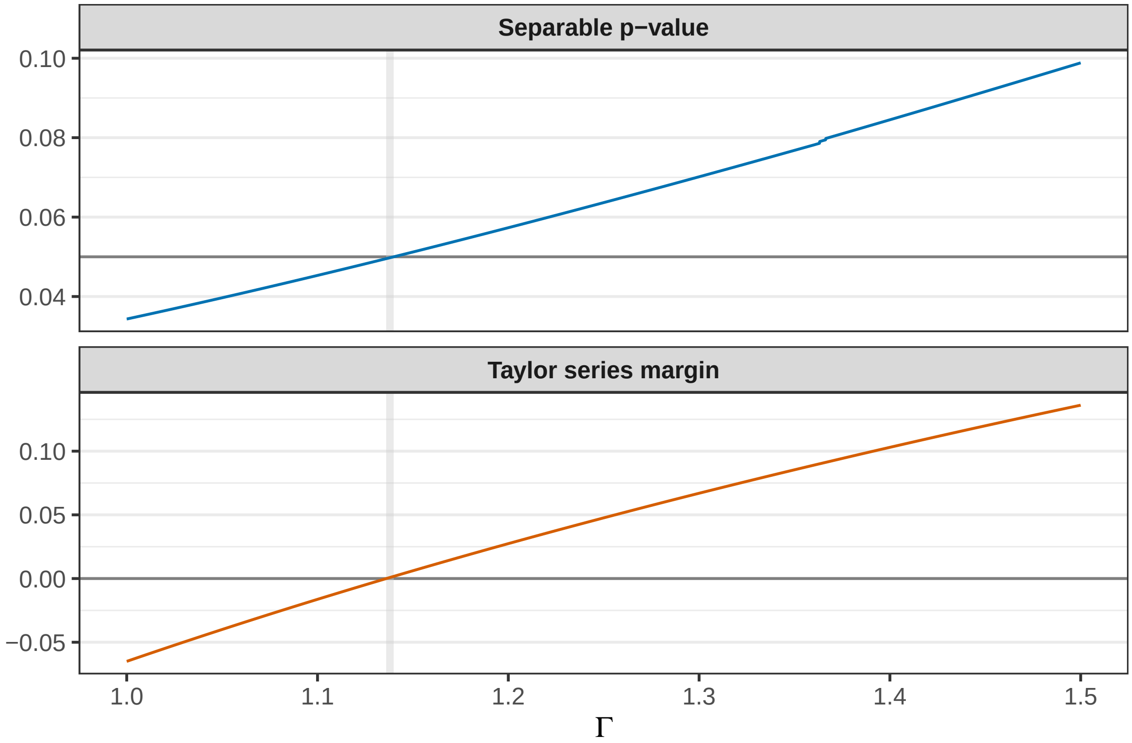

Figure 3: Sensitivity analysis for the sharp null of no individual treatment effect for all units. The top panel shows upper one-sided p-values under the separable approximation across values of Γ; the bottom panel reports, across the same values of Γ, the corresponding Taylor series margin, with the zero line marking the rejection boundary. The shaded vertical band highlights values of Γ for which the separable approximation and the Taylor series approximation lead to different rejection decisions.

The results in the shaded vertical band — corresponding to values of Γ for which the two approaches yield different conclusions (between 1.1361 and 1.1399) — show that the two methods diverge in exactly the way one would expect. Over this range of Γ values, the Taylor series margin lies just above zero, indicating no rejection of the sharp null, while the separable approximation yields p-values just below 0.05, indicating rejection. This pattern reflects the relative conservatism of the Taylor series approximation, since rejection under the Taylor series approach implies rejection under the separable approach, but not vice versa.

Both approaches to sensitivity analysis rely on multiple approximations. For example, the separable approximation is used to approximate the worst-case configuration of hidden bias $\boldsymbol{u} \in \mathcal{U}_+$, and inference additionally relies on a Normal approximation to the null randomization distribution of the test statistic.

If the exact randomization distribution were used instead, the resulting sensitivity conclusions could differ for certain values of Γ. In principle, one could also assess sensitivity at a given Γ using the exact randomization distribution. For each configuration of hidden bias $\boldsymbol{u} \in \mathcal{U}_{+}$, one would derive from (3) the implied probability distribution on Ω, combine that distribution with the previously defined `exact_sharp_null_dist` to compute the corresponding p-value, and take the largest of these p-values as the worst-case p-value (Rosenbaum and Krieger, 1990). A single configuration, such as the separable choice, gives only a lower bound on the worst-case p-value. Our study is small enough to compute the p-values for all 1,152 configurations in $\mathcal{U}_{+}$. The exact worst-case p-value first exceeds 0.05 at $\Gamma \approx 1.12$, compared with about 1.14 under the Normal approximation, so with only 12 matched sets the Normal approximation somewhat overstates the robustness of our conclusion.

2.4.2 How Do Inferences under the Weak Framework Change under These Departures?

Compared with tests of the sharp null hypothesis of no effect, constructing valid tests of an analogous weak null hypothesis is more challenging. A sharp null hypothesis specifies all missing potential outcomes, so we need only consider p-values over configurations of $\boldsymbol{u} \in \mathcal{U}_{+}$, rather than also over multiple configurations of potential outcomes. By contrast, a weak null permits many possible configurations of potential outcomes, so identifying the worst-case p-value requires optimization over both $\boldsymbol{u}$ and the potential outcomes consistent with the null. With binary outcomes, this joint optimization can be solved by integer programming under a Normal approximation (Fogarty et al., 2017). In matched-pair designs, the joint optimization is unnecessary because the worst-case expectation computed under the sharp null also bounds the expectation under the weak null, so a studentized sensitivity analysis is valid for the weak null in large samples (Fogarty, 2020a).

Fogarty (2023) proposes a more generally applicable approach, which avoids the need for an explicit search for the worst-case p-value over all configurations of $\boldsymbol{u}$ and potential outcomes consistent with the null hypothesis about τ. To build intuition for this approach, note that the following inverse probability weighted (IPW) version of the Difference-in-Means is unbiased for the ATE, τ:

$$\sum_{s=1}^{S} (n_s/n) \left(\frac{1}{|\Omega_s|} \right) \left(\frac{\hat{\tau}_s}{p(\boldsymbol{z}_s)} \right), \tag{5}$$

where $p(\boldsymbol{z}_s)$ denotes the probability of assignment $\boldsymbol{z}_s$ among all treatment assignments in set s that hold fixed the realized number of treated units, with $|\Omega_s|$ denoting, as defined earlier, the total number of such assignments. Both $\hat{\tau}_s$ and $p(\boldsymbol{z}_s)$ are evaluated at the same realized treatment assignment $\boldsymbol{z}_s$ within matched set s. In other words, $p(\boldsymbol{z}_s)$ denotes the probability of the assignment under which the treated and control outcomes in set s — and hence the Difference-in-Means $\hat{\tau}_s$ — are realized.

When $\Gamma = 1$, this probability $p(\boldsymbol{z}_s)$ is $1/|\Omega_s|$, and the IPW Difference-in-Means reduces to the usual Difference-in-Means, $\hat{\tau} = \sum_{s=1}^{S}(n_s/n)\hat{\tau}_s$. When $\Gamma > 1$, however, $p(\boldsymbol{z}_s)$ is no longer known exactly. In matched designs that may include sets with a single treated unit and many controls as well as sets with a single control unit and many treated units, Γ restricts the assignment probability to lie within bounds consistent with the specified degree of departure from as-if randomization:

$$\frac{1}{\Gamma(n_s-1)+1} \leq p(\boldsymbol{z}_s) \leq \frac{\Gamma}{(n_s-1)+\Gamma}, \tag{6}$$

for all $\boldsymbol{z}_s \in \Omega_s$ and for all matched sets s.

Since the IPW Difference-in-Means in (5) cannot be directly computed when $\Gamma > 1$, Fogarty (2023) instead uses the bounds in (6) to construct a worst-case version of this IPW Difference-in-Means. For a test of the weak null hypothesis $H_0\colon \tau = \tau_0$ against the alternative of a larger ATE, the worst-case IPW Difference-in-Means is

$$T_\Gamma(\tau_0) = \sum_{s=1}^{S}(n_s/n)\left(\frac{1}{|\Omega_s|}\right)\left(\frac{\hat{\tau}_s-\tau_0}{\tilde{p}_s}\right), \quad \text{where} \quad \tilde{p}_s = \begin{cases} \dfrac{\Gamma}{(n_s-1)+\Gamma} & \text{if } \hat{\tau}_s \geq \tau_0, \\[2ex] \dfrac{1}{\Gamma(n_s-1)+1} & \text{if } \hat{\tau}_s < \tau_0. \end{cases} \tag{7}$$

That is, the statistic replaces the unknown probability $p(\boldsymbol{z}_s)$ in (5) with whichever bound in (6) works against rejecting the null. Each set's Difference-in-Means is first centered at the null ATE. When the centered value $\hat{\tau}_s - \tau_0$ is positive, and so counts as evidence against the null, dividing it by the largest probability that Γ allows makes the set's contribution as small as possible. When the centered value is negative, dividing it by the smallest probability makes the set's contribution as negative as possible. When testing the null against the alternative of a smaller ATE, the two substitutions are reversed. At $\Gamma = 1$, both bounds equal $1/|\Omega_s|$, and $T_\Gamma(\tau_0)$ reduces to $\hat{\tau} - \tau_0$.

Technically, Fogarty (2023, Section 4.1) works with a rescaled version of this statistic, $\tilde{K}_{\Gamma}^{(\tau_0)}$, which is exactly twice $T_{\Gamma}(\tau_0)$. The rescaling does not change the test, because the standard error used to standardize the statistic is multiplied by the same factor of 2. The worst-case IPW form of the statistic comes from the preprint version of Fogarty (2023), which describes each matched set's quantity $(1/|\Omega_s|)(\hat{\tau}_s - \tau_0)/\tilde{p}_s$ in (7) as a worst-case inverse probability weighted estimator (Fogarty, 2020b, Sections 5.2–5.3).

To see the value of this procedure, consider testing a null hypothesis about the ATE against the alternative of a larger ATE. Fogarty (2023) shows that, for any $\Gamma \geq 1$, the expected value of this worst-case IPW Difference-in-Means — defined under whatever the true distribution on Ω consistent with that Γ happens to be — is always less than or equal to zero when the null is true, because the statistic is centered at τ_0. Conversely, when testing against the alternative of a smaller ATE, this expectation is greater than or equal to zero when the null is true. Importantly, these properties hold for all possible values of $\boldsymbol{u}$ and all configurations of potential outcomes that are consistent with the null hypothesis.

How does this property of the expected value ensure a valid test in sufficiently large studies? Focusing on the case of testing against a larger ATE, the intuition is straightforward. If the expected value of the worst-case IPW Difference-in-Means is less than or equal to zero when the null is true, then the worst-case IPW Difference-in-Means tends to fall at or below zero rather than above it. In other words, the procedure intentionally "tilts" the worst-case IPW Difference-in-Means in the direction opposite the alternative, making the procedure conservative.

In addition to this property of the expected value, a second key ingredient for valid hypothesis testing concerns variance estimation. For any fixed $\Gamma \geq 1$, an analogue of the variance estimator from Fogarty (2018) (discussed in Section 2.3.2) is conservative for the true variance of the worst-case IPW Difference-in-Means, in that the estimator's expected value is at least that variance and, as the number of matched sets grows, the estimator's ratio to that variance converges to a value of at least 1 (Fogarty, 2023, Lemma 2). As a result, in large studies, using this variance estimator tends to reduce the magnitude of the standardized test statistic — the worst-case IPW Difference-in-Means, which is already centered at the null ATE, divided by the square root of the estimated variance — relative to what would be obtained using the true (but unknown) variance.

Together, these two properties work in the same direction:

- The expected value is at or below zero, keeping the center of the distribution at or shifted away from the upper tail, and
- the variance estimate tends to be too large, which pulls the standardized statistic closer to zero.

As a result, when the null is true, the probability that the standardized statistic falls in the upper tail of the standard Normal distribution — that is, the probability of falsely rejecting the null — remains at or below the nominal significance level. Hence, the test is conservative (valid) asymptotically, as the number of matched sets grows (Fogarty, 2023, Theorem 2). With few matched sets, as in our application with 12, the Normal approximation and the variance estimator can both be poor, so the test can over-reject, and readers should regard the resulting p-values as approximate. Analogous reasoning applies when testing against the alternative of a smaller effect. When the null hypothesis is true, the probability that the standardized value falls in the lower tail of the standard Normal distribution also stays at or below the nominal significance level.

To implement this approach from Fogarty (2023), we first source from the GitHub repository an `R` function that, for a fixed $\Gamma \geq 1$, computes the quantity $(1/|\Omega_s|)(\hat{\tau}_s - \tau_0)/\tilde{p}_s$ in (7) for a single matched set s.

```r
# Load the worst_case_IPW() function from GitHub repo
source(
  paste0(
    base_url, "/R/worst_case_IPW.R"
  )
)
```

We now use this function to calculate the worst-case IPW version of the Difference-in-Means within each matched set. To form the overall statistic $T_{\Gamma}(\tau_0)$ in (7), we average the set-specific values using weights equal to each set's share of the total number of units, n_s/n. Because `worst_case_IPW()` already centers each set's Difference-in-Means at the null ATE before applying the bounds, no further centering is needed. Finally, we standardize this statistic using the conservative variance estimator of Fogarty (2018) from Section 2.3.2, and compare the resulting standardized test statistic to the standard Normal distribution to obtain p-values. For the rescaled statistic $\tilde{K}_{\Gamma}^{(\tau_0)}$, Fogarty (2023, p. 2202) uses different values in the projection step of the variance estimator. Any fixed choice of these values gives a conservative variance estimator (Fogarty, 2018, Proposition 1), and on our data the two choices give nearly identical sensitivity values (1.1756 and 1.1753 on our grid).

```r
# One-sided sensitivity analysis for the weak null (ATE = 0)

# Null and alternative for the weak-null test
null_ATE    <- 0                  # Null: ATE = 0

alternative <- "greater"          # Alt: ATE > 0 (upper-tail test)

# Data frame to store results for each Gamma
weak_sens_df <- data.frame(
  Gamma   = Gamma_vals,
  p_value = NA_real_
)

for (g in Gamma_vals) { # Loop over Gamma values

  # Worst-case IPW estimate within each matched set at Gamma = g
  set_stats <- data_matched |>
    group_by(
      fm
    ) |>
        summarize(
      n      = n(),                       # Set size n_s
      weight = n / nrow(data_matched),     # Weight n_s / n
      est    = worst_case_IPW(             # Worst-case IPW in set s
        z           = UN,                 # Treatment indicator
        y           = ldur,               # Outcome
        Gamma       = g,                  # Sensitivity parameter (Gamma)
        tau_h       = null_ATE,           # Weak null: ATE = 0
        alternative = alternative         # Direction of alternative
      ),
      .groups = "drop"
    )

  # Overall worst-case IPW statistic (weighted average across sets)
  wc_ipw_stat <- sum(
    set_stats$weight * set_stats$est
  )

  # Variance estimate for the worst-case IPW statistic
  var_hat <- fine_strat_var_est(
    strat_ns   = set_stats$n,
    strat_ests = set_stats$est
  )
  # Standardized worst-case IPW statistic, centered at the null
  se_hat   <- sqrt(
    var_hat
  )

  std_stat <- wc_ipw_stat / se_hat

  # One-sided p-value, determined by the alternative
  if (alternative == "greater") {
    # Upper-tail test: ATE > 0
    p_val <- 1 - pnorm(std_stat)
  } else if (alternative == "less") {
    # Lower-tail test: ATE < 0
    p_val <- pnorm(std_stat)
  } else {
    stop("Only one-sided alternatives ('greater' or 'less').")
  }

  # Store result for this Gamma
  weak_sens_df$p_value[weak_sens_df$Gamma == g] <- p_val
}

# Sensitivity value: smallest Gamma where one-sided test no longer rejects
weak_sens_value <- min(
  weak_sens_df$Gamma[weak_sens_df$p_value >= alpha]
)
```

Figure 4 plots the upper one-sided p-values corresponding to Γ values from 1 to 1.5.

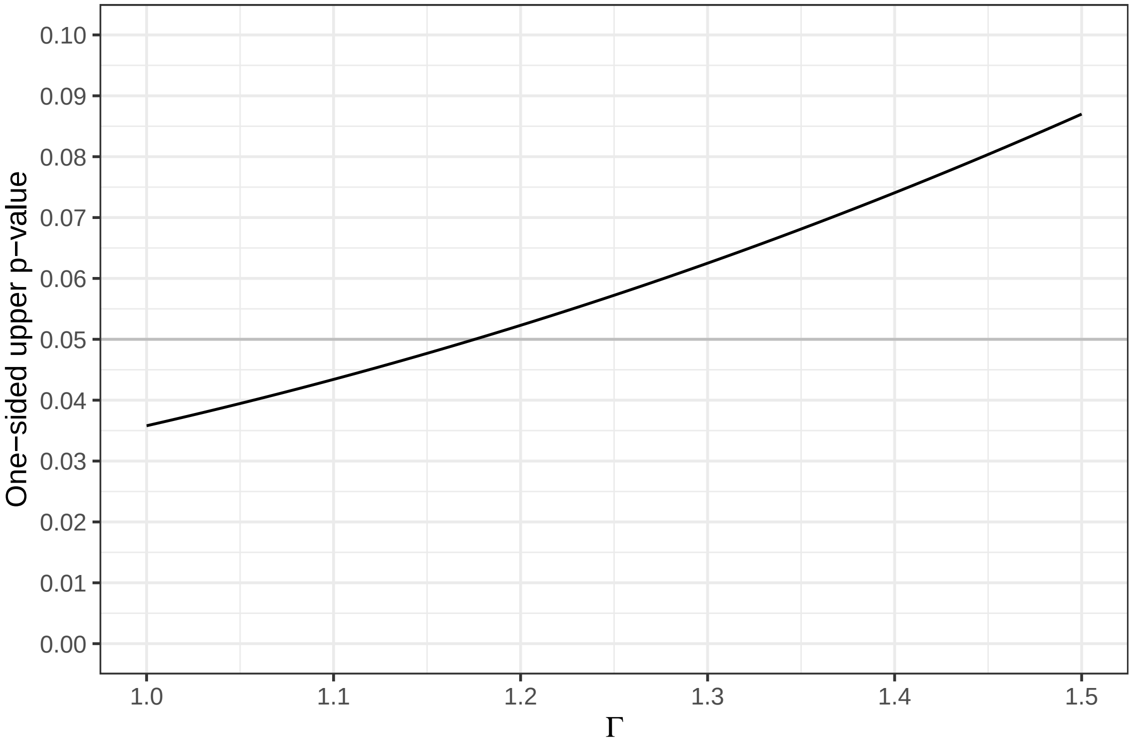

Figure 4: Upper one-sided p-values for the weak null hypothesis of zero average effect across values of the sensitivity parameter Γ.

Under as-if randomization ($\Gamma = 1$), we find evidence of a positive average effect; the p-value at $\Gamma = 1$ in Figure 4 is 0.036, rather than the 0.037 reported in Section 2.3.2, because the sensitivity analysis estimates the variance with the estimator of Fogarty (2018) instead of `hybrid_p`. This evidence is sensitive to departures from as-if randomization. Our qualitative conclusion about the null hypothesis that the ATE equals zero changes at $\Gamma \approx 1.18$ (1.1756 on our grid). This sensitivity value is similar to the sensitivity value for the sharp null of no effect, but the two analyses use different test statistics, variance calculations, and approximations, so readers should not take the small difference between the two sensitivity values as evidence that the weak-null conclusion is more robust. With only 12 matched sets, both analyses rest on fragile Normal approximations, and we regard both sensitivity values as rough.

This sensitivity analysis guarantees (in large studies) validity across *all* configurations of potential outcomes consistent with the weak null. The price of this guarantee is conservativeness. If effects are in fact constant, the size of the test is strictly below α in designs other than matched pairs (Fogarty, 2023, Theorem 2). By contrast, Fogarty (2023) also describes, and recommends, an alternative sensitivity analysis that is asymptotically exact when effects are constant but is valid for the weak null only under an additional condition that jointly restricts the potential outcomes and the pattern of hidden bias, a condition that researchers may regard as plausible (Fogarty, 2023, Theorem 1 and Condition 1). The alternative sensitivity analysis can produce smaller p-values, but at the cost of forfeiting validity for certain (possibly unrealistic) combinations of potential outcomes and hidden bias.

3 Conclusion

We have now completed the full matching pipeline. We began by constructing matched sets and then conducted inference under the as-if randomization assumption, treating the matched design as a collection of completely randomized experiments within blocks. We examined inference under both the sharp and weak frameworks — corresponding, respectively, to unit-level causal effects and average effects — and then assessed the sensitivity of these inferences to potential violations of the as-if randomization assumption. The embedded code, accompanied by extensive comments and illustrated through the running example from Gilligan and Sergenti (2008), is intended to be readily adapted by practitioners to their own datasets.

3.1 Limitations and Related Topics Beyond Our Scope

For clarity and focus, we have excluded several important topics, though we have pointed to relevant references where appropriate. Some of these omissions are not specific to matching or observational studies. In particular, we have not addressed imperfect compliance with treatment assignment, missing outcomes, clustered (rather than individual-level) assignment, or interference settings in which a subject's outcome depends on others' treatment assignments.

Our analysis has focused on inference for two causal targets only: a constant, additive treatment effect and the average treatment effect. We have not considered alternative effect models, such as dilated effects (Rosenbaum, 1999), multiplicative effects, or Tobit effects (Rosenbaum, 2010, pp. 46–48). Throughout, we have treated causal targets as fixed quantities (over the set of treatment assignments consistent with the matched design), rather than as random variables. As a result, we have excluded attributable effects (Rosenbaum, 2001, 2003; Hansen and Bowers, 2009) and the average treatment effect on the treated (Sekhon and Shem-Tov, 2021). We have also set aside methods for joint inference on sharp and weak causal hypotheses (Wu and Ding, 2021; Cohen and Fogarty, 2022) (see also Ding, 2017), as well as strategies that improve the power of hypothesis tests by rescaling outcomes or choosing alternative test statistics, including regression-assisted approaches (Lin, 2013; Cohen and Fogarty, 2024; Guo and Basse, 2023).

Several omissions are specific to matching methodology itself:

- First, we have not covered a range of **alternative matching and balancing approaches**, including nonbipartite matching for multivalued treatments (Lu et al., 2001, 2011; Rosenbaum, 2010, pp. 207–221), template matching (Silber et al., 2014), multilevel matching (Zubizarreta and Keele, 2017; Pimentel et al., 2018), risk-set matching (Li et al., 2001), cardinality matching (Zubizarreta et al., 2014), coarsened exact matching (Iacus et al., 2012), exterior matching (Rosenbaum and Silber, 2013), or generalized full matching (Sävje et al., 2021), among others. We have further assumed that researchers specify *ex ante* the set of covariates to be balanced, thereby excluding screening procedures that identify additional covariates for adjustment based on observed imbalances (Cochran, 1965), as in tapered matching approaches (Daniel et al., 2008; Yu et al., 2021). Finally, we have not considered balance constraints such as fine and near-fine balance (Rosenbaum et al., 2007; Yang et al., 2012; Yu, 2023; Pimentel et al., 2015a,b), which enforce exact or nearly exact equality of the marginal distributions of selected nominal covariates.
- Second, we have not discussed the use of **prognostic covariates** identified using pilot or external data. Such approaches can substantially improve efficiency by prioritizing covariates that strongly predict potential outcomes. For theoretical foundations, see Hansen (2008a), with complementary theoretical results and practical insights in Sales et al. (2018).
- Third, we have considered only matching strategies aimed at justifying as-if randomization, not those designed explicitly to improve **design sensitivity** — that is, robustness to moderate violations of as-if randomization (see Rosenbaum, 2010, Part III, pp. 257–311). Beyond match construction, researchers can also enhance design sensitivity by selecting particular test statistics, among other strategies (Rosenbaum, 2004; Heller et al., 2009; Hsu et al., 2013; Small et al., 2013).
- Fourth, we have excluded **extensions to covariate balance testing** beyond the framework developed in Hansen and Bowers (2008), including subsequent contributions by Gagnon-Bartsch and Shem-Tov (2019) and Branson (2021), as well as related approaches based on the stepwise

intersection–union principle (Hansen and Sales, 2015).

- Fifth, we have ignored **residual imbalance on observed covariates**. Specifically, we have proceeded under the assumption that matched set membership can justify as-if randomization, even when small imbalances remain within matched sets. Under this approach, concerns about residual imbalance are absorbed into the sensitivity analysis for hidden bias. This approach is not entirely satisfying, however, because imbalances on observed covariates are not, in fact, hidden. Several methods therefore address such imbalances directly, either in approaches targeting individual effects (Rosenbaum, 1988; Pimentel and Huang, 2024; Chen et al., 2023; Heng et al., 2025) or average effects (Zhu et al., 2025).
- Sixth, we have not incorporated into the overall pipeline an explicit approach for defining an **interpretable study population** in matched designs. Common practice excludes units based on limited overlap in the estimated propensity score (Crump et al., 2009; King and Zeng, 2006), but study populations defined in this way can be difficult to characterize substantively. Hence, alternative approaches define the study population directly in terms of a small number of covariates. For example, Traskin and Small (2011) use a classification tree to define a study population that approximates one defined by trimming on the estimated propensity score, while Fogarty et al. (2016) use discrete optimization to define a maximal, rectangular region of covariate space with adequate overlap on key covariates.
- Seventh, we have not considered **quasi-experimental devices** (Campbell, 1957; Campbell and Stanley, 1963; Shadish et al., 2002), specifically how matched designs can leverage such devices (see Rosenbaum, 2025, Part IV). Two prominent quasi-experimental devices are an additional control group (Rosenbaum, 1987b) and outcomes with known effects (Rosenbaum, 1989), each of which has a range of valuable applications. In the context of evidence factors (Rosenbaum, 2021), one use of a second control group is to yield stronger causal evidence that is less susceptible to hidden confounding (Rosenbaum, 2023a). For known effects, one illustrative use is to support tests for hidden confounding that inform sensitivity analyses by ruling out certain violations of as-if randomization (Rosenbaum, 2023b).
- Eighth and finally, we have not addressed ***Z*-dependence** (Pashley et al., 2021; Pimentel and Huang, 2024; Pimentel and Yu, 2024). Our exposition has relied on an analogy to Rubin's framework of "assignment to treatment group on the basis of a covariate" (Rubin, 1977), where matched

set membership plays the role of the covariate. This analogy treats set membership as fixed, but because matching uses the realized treatment assignment, other assignments can produce different matched sets. In our application, permuting treatment within any single matched set and re-running our matching code produces a different matched structure. Assignments that would have produced a different matched structure do not belong in the reference distribution, and the standard justification for permuting treatments within sets requires exact matching on all matching variables (Pashley et al., 2021, Section 5). In the simulations of Pimentel and Huang (2024) and Pimentel and Yu (2024), propensity score calipers reduced, but did not always remove, the resulting excess Type I error. We thank Samuel Pimentel for helpful discussions of $\boldsymbol{Z}$-dependence.

Despite these limitations, the pipeline and design-based perspective developed here provide a coherent framework for understanding how matched designs support causal inference. The emphasis on design-based foundations clarifies the logic underlying the construction of matched sets and the subsequent steps of inference and sensitivity analysis under both weak and sharp frameworks. We present this work not as exhaustive, but as a foundation on which researchers can build when incorporating more specialized methods tailored to their substantive and inferential goals.

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